

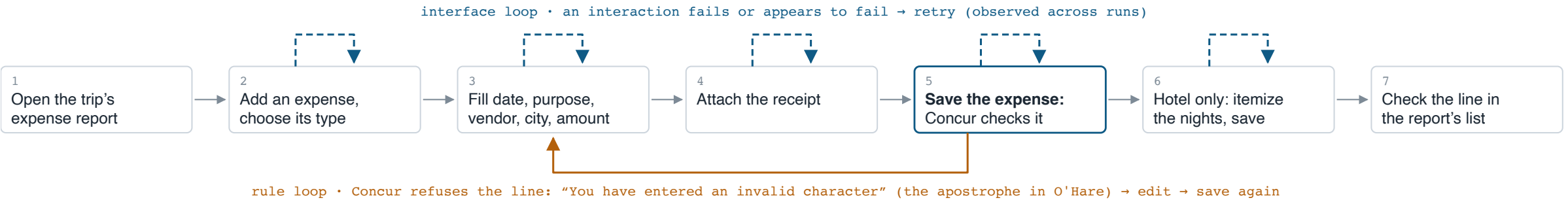
Agent Acceleration in a Stable Environment

Learning from repeated tasks to make later ones faster and cheaper

October 2026

Example: a web agent that files expense reports in Concur

Each task is a new expense report on the same Concur site, filed line by line; the agent never submits it. Two kinds of loop cost it steps, a rule of the site it does not know and an interface it does not handle, and both came back on later reports until someone wrote the fix down.



A RULE OF THE SITE · TRIP A

Expense Type *

53103-US LOCAL TRANSPORTATION

Transaction Date *

08/10/2026

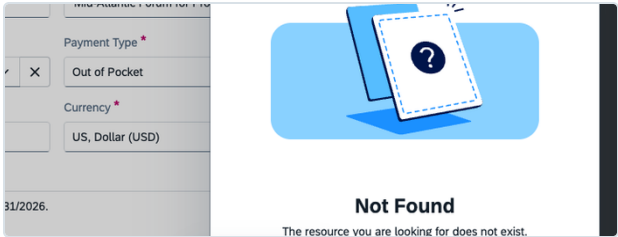
Business Purpose *

Uber, O'Hare Airport to hotel, MOSS 2026 Chica

Vendor Name *

Concur refuses the apostrophe in “O'Hare”. It came back twice on 14 September; that evening the rule went into the agent's prompt by hand, and it has not come back.

THE INTERFACE · TRIP R



The line had been saved; the dialog said otherwise: 60 more saves, \$4.20. Fixed in code: a save is judged by the line in the list.

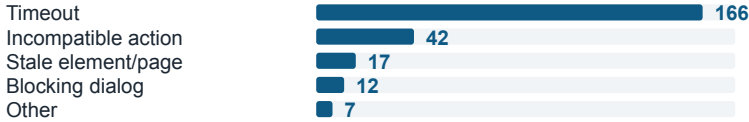
Runs showing Concur errors



A run may appear in more than one category.

Action failures

244 failed actions out of 11,103 attempts (2.2%)



Batch-run failures



LEARNING A RULE → a note: “no apostrophes in Business Purpose”, in the prompt, the memory or a store of site facts

LEARNING THE INTERFACE → code: “a save worked if the line is in the list”, as a skill, a tool or a harness change

OUTSIDE THE AGENT → engineering and reruns: counted in time and cost, not against success; not learning

Five difficulties, grouped by the cost they affect

COST PER TASK AFTER LEARNING · $\bar{v}(m')$

1 · Fewer steps do not always save time

BEFORE

AFTER

10.7 steps · 215 s

10.2 steps · 395 s

one step

extra reading and thinking in each step

length = seconds per task · per step 20.1 → 38.7 s (calc.)

Each step got longer: judge a report by its seconds, not its steps

Drawn from WebCoach (Liu et al., 2025). Also: ReasoningBank (Ouyang et al., 2026), fewer steps, more tokens

2 · Which observations can we safely skip?

Choose type: Hotel

check the form

fill dates

save

✓

one step more, always right

skip: as last time

save

✓ rule still holds

✗ it changed: rework

schematic

Skipping a check saves a step only while the rule still holds

Seen in: DRAFT (Qu et al., 2025), learns tool conditions by trying · SKILL.nb (Hattami et al., 2026), checks before reuse

3 · A trade-off between generalization and cost per use

generalizes ↑

cheap and general?

note in every prompt

note in memory

skill: steps + script

replayed script

cost per use →

schematic

Seen in: Metis (Dai et al., 2026), keeps both text and code · ActionEngine (Zhong et al., 2026), turns a site map into programs

COST OF LEARNING · C_{learn}

4 · Finding the cause behind an error message

Error: over the meal limit

possible causes, of different kinds, checked one by one

reasoning wrong type?

rule limit changed?

interaction wrong field?

tool amount misread?

...?

the cause may be found late, or not at all

cost of checking so far →

schematic

Each possible cause costs a check; the right one may come late, or never

Seen in: HarnessFix (Chen et al., 2026), finds the faulty step · ESPO (L. Liu et al., 2026), sorts errors first

5 · Learning and testing can cost more than they save

total cost ↑

learning and testing are paid first

break-even n^*

break-even, more testing

repair after the site changes

no learning more testing with learning

number of reports filed →

schematic

Learning pays off only if enough reports share its cost

Seen in: AI Agents That Matter (Kapoor et al., 2025), late break-even · SICA (Robeyns et al., 2025), large bill, small saving

REFERENCES

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• Dai, Z., He, S., Li, H., et al. (2026). Metis: Bridging Text and Code Memory for Self-Evolving Agents. *arXiv preprint*.

• Hattami, A. E., Chapados, N., & Pal, C. (2026). SKILL.nb: Selective Formalization and Gated Execution for Durable Agent Workflows. *FAGEN@ICML 2026 Poster (ICML 2026 workshop, venueid ICML.cc/2026/Workshop/FAGEN)*.

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• Qu, C., Dai, S., Wei, X., et al. (2025). From Exploration to Mastery: Enabling LLMs to Master Tools via Self-Driven Interactions. *ICLR 2025 (International Conference on Learning Representations 2025)*.

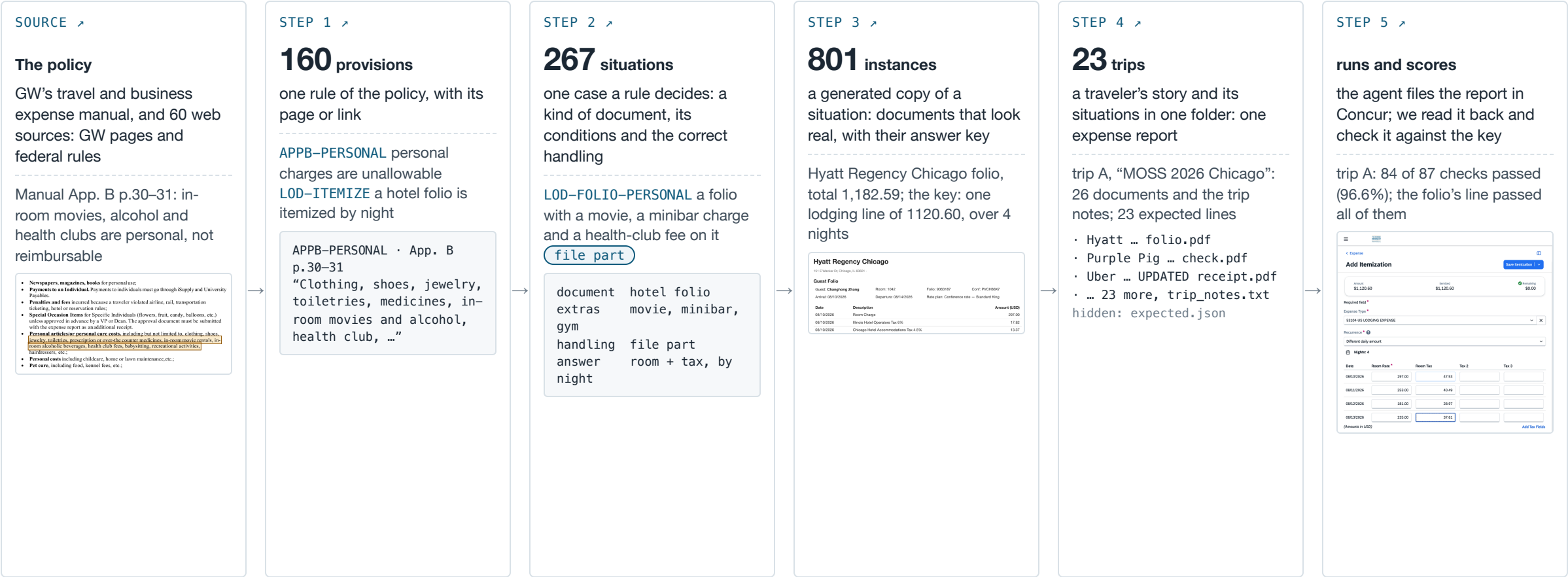
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• Zhong, H., Faisal, F., França, L., et al. (2026). ActionEngine: From Reactive to Programmatic Web Agents via State Machine Memory. *arXiv preprint*.

How we generate the ExpenseAI evaluation data

We break GW’s expense policy into rules, write each case a rule decides as a situation, generate documents and an answer key for it, bundle situations into expense reports, and score each report the agent files against its key.

FOLLOWED THROUGH THIS SECTION: ONE HOTEL FOLIO WITH PERSONAL CHARGES ON IT, FROM TRIP A (CHICAGO)

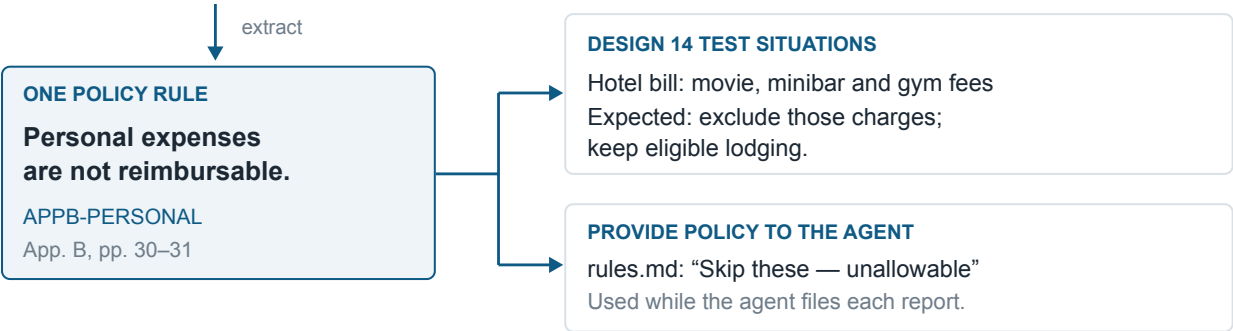


Provisions: the policy as 160 rules, each with its page or link

A provision is one rule of the policy, recorded with its page or link; the 149 rules an agent can act on while filing are each tested by at least one situation.

WHAT THE MANUAL SAYS · APPENDIX B, P.30–31

- **Newspapers, magazines, books** for personal use;
- **Payments to an Individual.** Payments to individuals must go through iSupply and University Payables.
- **Penalties and fees** incurred because a traveler violated airline, rail, transportation ticketing, hotel or reservation rules;
- **Special Occasion Items** for Specific Individuals (flowers, fruit, candy, balloons, etc.) unless approved in advance by a VP or Dean. The approval document must be submitted with the expense report as an additional receipt.
- **Personal articles/or personal care costs.** including but not limited to, clothing, shoes, jewelry, toiletries, prescription or over-the counter medicines, in-room movie rentals, in-room alcoholic beverages, health club fees, babysitting, recreational activities, hairdressers, etc.;
- **Personal costs** including childcare, home or lawn maintenance, etc.;
- **Pet care**, including food, kennel fees, etc.;



KIND 1 · WHERE A PROVISION COMES FROM

- 83 the manual** (rev. 2024-02-21)
[LOD-ITEMIZE](#) p.12–13: a hotel folio is itemized by night, room rate and tax; personal charges on it are left out
- 77 60 web sources:** GW pages and federal rules, for expenses outside travel
[NC-SELF-PAID](#): subscriptions, supplies, cloud services and dues paid out of pocket are filed, with a notice that GW prefers its P-Card or iBuy+

KIND 2 · CAN AN AGENT ACT ON IT WHILE FILING?

- 149 yes:** it decides how a document is filed
[APPB-PERSONAL](#), on the left
- 11 no:** it acts before or after filing
[ADVANCE](#) p.18–19: a travel advance is reconciled against the report, not filed from a receipt

Listing the 11 keeps “the whole policy is covered” honest: it means every rule an agent can act on, not only the convenient ones.

Situation: a document, its conditions and its correct handling

A situation is one case the policy decides, apart from any trip; the same document under other conditions is another situation.

ONE SITUATION, AS WE WROTE IT

SITUATION	LOD-FOLIO-PERSONAL family: lodging
DOCUMENT	a hotel folio with an in-room movie, a minibar charge and a health-club fee on it
CONDITIONS	a trip to a US city; any traveler; any funding
HANDLING	file part the room and its tax only, itemized by night
ANSWER KEY	one lodging line, night by night; the folio's total must not appear as a line
BASIS	manual p.12–13 and App. B p.30–31; one of our own hotel folios
FORM WORK	complex: the nights are entered on a second screen

Hyatt Regency Chicago		
151 E Wacker Dr, Chicago, IL 60601		
Guest Folio		
Guest: Changhong Zhang	Room: 1042	Folio: 9063187
Arrival: 08/10/2026	Departure: 08/14/2026	Conf: P/CH88X7
Rate plan: Conference rate — Standard King		
Date	Description	Amount (USD)
08/10/2026	Room Charge	297.00
08/10/2026	Illinois Hotel Operators Tax 6%	17.82
08/10/2026	Chicago Hotel Accommodations Tax 4.5%	13.37
08/10/2026	Cook County Hotel Tax 1%	2.97
08/10/2026	MPEA Hotel Tax 2.5%	7.43
08/10/2026	Sports Facilities Tax 2%	5.94
08/11/2026	Room Charge	253.00
08/11/2026	Illinois Hotel Operators Tax 6%	15.18
08/11/2026	Chicago Hotel Accommodations Tax 4.5%	11.39
08/11/2026	Cook County Hotel Tax 1%	2.53
08/11/2026	MPEA Hotel Tax 2.5%	6.33
08/11/2026	Sports Facilities Tax 2%	5.06
08/11/2026	In-Room Movie	14.99
08/12/2026	Room Charge	181.00
08/12/2026	Illinois Hotel Operators Tax 6%	10.86
08/12/2026	Chicago Hotel Accommodations Tax 4.5%	8.15
08/12/2026	Cook County Hotel Tax 1%	1.81
08/12/2026	MPEA Hotel Tax 2.5%	4.53
08/12/2026	Sports Facilities Tax 2%	3.62
08/12/2026	Minibar	22.00
08/13/2026	Room Charge	235.00
08/13/2026	Illinois Hotel Operators Tax 6%	14.10
08/13/2026	Chicago Hotel Accommodations Tax 4.5%	10.58
08/13/2026	Cook County Hotel Tax 1%	2.35
08/13/2026	MPEA Hotel Tax 2.5%	5.88
08/13/2026	Sports Facilities Tax 2%	4.70
08/13/2026	Health Club / Fitness Center	25.00
Total		1182.99 USD
Payment — Visa XXXX4417		-1182.99 USD
Balance		0.00 USD

one generated copy (trip A): the three personal charges and the total are highlighted

THE SAME DOCUMENT, OTHER CONDITIONS

The Westin Copley Place, Boston		
10 Huntington Ave, Boston, MA 02116 -		
Guest Folio		
Guest: Changhong Zhang	Room: 1846	Folio: 4713770
Arrival: 08/24/2026	Departure: 08/27/2026	Conf: N8KR8QRM
Rate plan: Conference rate — Standard King		
Date	Description	Amount (USD)

a folio with a high-speed internet charge on it

SITUATION	LOD-FOLIO-INTERNET
CONDITION	paid from department funds
HANDLING	file
ANSWER KEY	one lodging line, the internet included

SITUATION	LOD-FOLIO-INTERNET-GRANT
CONDITION	paid from a federal research grant
HANDLING	file + flag
ANSWER KEY	one lodging line, and a flag: internet is unallowable on the grant

267 situations: 169 in travel, 98 outside it. None is tied to a trip: each can be generated for any city, dates, traveler and funding its conditions allow.

Five ways a document can be handled

Every situation has one correct handling; 196 of the 267 ask for more than copying the receipt, and those are where the agent fails: a situation needing a flag passed 63% of its tests, plain filing 96%.

file 71 situations

one expense line, as the receipt reads

UNITED

Your Flight Receipt - CHANGHONG ZHANG - 24AUG26

Confirmation Number
XN8PE2

Passenger Info
Name: CHANGHONG ZHANG

FLIGHT	DATE	DEPART	ARRIVE
UNITED 1664 (Economy J)	MON 24AUG	RONALD REAGAN WASHINGTON NATIONAL AIRPORT 07:00AM	BOSTON LOGAN INTERNATIONAL AIRPORT 08:31AM

AIR-COACH-DIRECT a coach ticket bought from the airline → one airfare line

passed 243 of 253 tests (96%)

A FLAG HAS A LEVEL • NON-TRAVEL DATA

notice: filed as it is, the reader is told something — supplies paid out of pocket, with GW's preferred P-Card named (NC-SELF-PAID)

needs review: a person must decide — a computer under \$5,000 bought by the traveler, filed under 52193, since GW IT orders computers

file + flag 97 situations

one line, and a note for a person to decide before the report is submitted

NEDS Society

101 Federal St, Suite 1000, Boston, MA 02110
(617) 555-5142
https://nedsociety.org
registration@nedsociety.org

THANK YOU!

Receipt Info
Date: 8/4/2026
Receipt Number: 3887364828
Customer: Changhong Zhang
Total: \$190.00

QUANTITY	LINE DESCRIPTION	LIST PRICE	SALE PRICE	TOTAL
1	Virtual Workshop on Thoughtworthy ML - online registration	\$190.00	\$190.00	\$190.00
			SUBTOTAL:	\$190.00
			TOTAL:	\$190.00

REG-VIRTUAL an online-only conference → a registration line, flagged

passed 84 of 134 tests (63%)

separate out 6 situations

one receipt becomes two lines, on two accounts

The Purple Pig

444 N Michigan Ave, Chicago, IL 60611

08/12/2026 1:12 PM Check #70637
Server: Camila Table: 6 Guests: 3

1	Pasta Primavera	22.00
1	Grilled Salmon	28.00
1	Fish Tacos	21.95
1	Old Fashioned	16.00
1	Old Fashioned	16.00
	Subtotal	103.95
	Sales Tax 10.25%	10.65
	Restaurant Tax 0.5%	0.52
	Tip	20.79
	TOTAL USD	135.91

MEAL-BUSINESS-ALCOHOL lunch with two cocktails → 94.07 business meal with the attendees, 41.84 to entertainment (52611), flagged

passed 7 of 10 tests (70%)

file part 15 situations

the allowable part is filed; the full total must not appear

Hyatt Regency Chicago

151 N Dearborn St, Chicago, IL 60611

Guest Folio

Arrival: 08/10/2026 Room: 1042 Rate: \$60.00 Tax: \$12.00
Departure: 08/14/2026 New York Conference with - Standard King

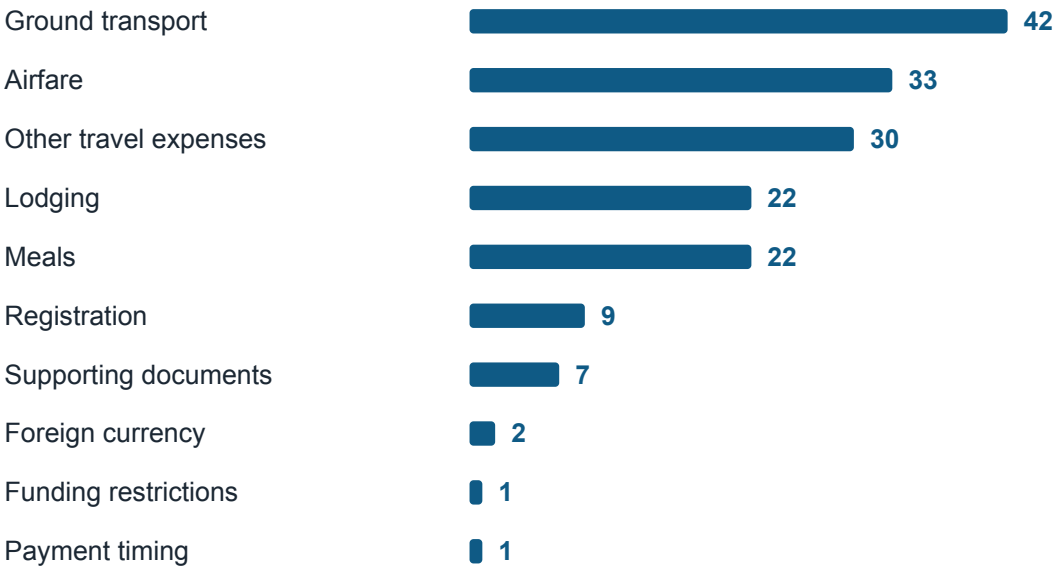
Date	Description	Amount (USD)
08/10/2026	Room Charge	60.00
08/10/2026	Room Hotel Operations Tax 6%	3.60
08/10/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/10/2026	Cook County Hotel Tax 1%	0.60
08/10/2026	MPIRA Hotel Tax 2.5%	1.50
08/10/2026	State Facilities Tax 2%	1.20
08/11/2026	Room Charge	60.00
08/11/2026	Room Hotel Operations Tax 6%	3.60
08/11/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/11/2026	Cook County Hotel Tax 1%	0.60
08/11/2026	MPIRA Hotel Tax 2.5%	1.50
08/11/2026	State Facilities Tax 2%	1.20
08/12/2026	Room Charge	60.00
08/12/2026	Room Hotel Operations Tax 6%	3.60
08/12/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/12/2026	Cook County Hotel Tax 1%	0.60
08/12/2026	MPIRA Hotel Tax 2.5%	1.50
08/12/2026	State Facilities Tax 2%	1.20
08/13/2026	Room Charge	60.00
08/13/2026	Room Hotel Operations Tax 6%	3.60
08/13/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/13/2026	Cook County Hotel Tax 1%	0.60
08/13/2026	MPIRA Hotel Tax 2.5%	1.50
08/13/2026	State Facilities Tax 2%	1.20
08/14/2026	Room Charge	60.00
08/14/2026	Room Hotel Operations Tax 6%	3.60
08/14/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/14/2026	Cook County Hotel Tax 1%	0.60
08/14/2026	MPIRA Hotel Tax 2.5%	1.50
08/14/2026	State Facilities Tax 2%	1.20
08/15/2026	Room Charge	60.00
08/15/2026	Room Hotel Operations Tax 6%	3.60
08/15/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/15/2026	Cook County Hotel Tax 1%	0.60
08/15/2026	MPIRA Hotel Tax 2.5%	1.50
08/15/2026	State Facilities Tax 2%	1.20
08/16/2026	Room Charge	60.00
08/16/2026	Room Hotel Operations Tax 6%	3.60
08/16/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/16/2026	Cook County Hotel Tax 1%	0.60
08/16/2026	MPIRA Hotel Tax 2.5%	1.50
08/16/2026	State Facilities Tax 2%	1.20
08/17/2026	Room Charge	60.00
08/17/2026	Room Hotel Operations Tax 6%	3.60
08/17/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/17/2026	Cook County Hotel Tax 1%	0.60
08/17/2026	MPIRA Hotel Tax 2.5%	1.50
08/17/2026	State Facilities Tax 2%	1.20
08/18/2026	Room Charge	60.00
08/18/2026	Room Hotel Operations Tax 6%	3.60
08/18/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/18/2026	Cook County Hotel Tax 1%	0.60
08/18/2026	MPIRA Hotel Tax 2.5%	1.50
08/18/2026	State Facilities Tax 2%	1.20
08/19/2026	Room Charge	60.00
08/19/2026	Room Hotel Operations Tax 6%	3.60
08/19/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/19/2026	Cook County Hotel Tax 1%	0.60
08/19/2026	MPIRA Hotel Tax 2.5%	1.50
08/19/2026	State Facilities Tax 2%	1.20
08/20/2026	Room Charge	60.00
08/20/2026	Room Hotel Operations Tax 6%	3.60
08/20/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/20/2026	Cook County Hotel Tax 1%	0.60
08/20/2026	MPIRA Hotel Tax 2.5%	1.50
08/20/2026	State Facilities Tax 2%	1.20
08/21/2026	Room Charge	60.00
08/21/2026	Room Hotel Operations Tax 6%	3.60
08/21/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/21/2026	Cook County Hotel Tax 1%	0.60
08/21/2026	MPIRA Hotel Tax 2.5%	1.50
08/21/2026	State Facilities Tax 2%	1.20
08/22/2026	Room Charge	60.00
08/22/2026	Room Hotel Operations Tax 6%	3.60
08/22/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/22/2026	Cook County Hotel Tax 1%	0.60
08/22/2026	MPIRA Hotel Tax 2.5%	1.50
08/22/2026	State Facilities Tax 2%	1.20
08/23/2026	Room Charge	60.00
08/23/2026	Room Hotel Operations Tax 6%	3.60
08/23/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/23/2026	Cook County Hotel Tax 1%	0.60
08/23/2026	MPIRA Hotel Tax 2.5%	1.50
08/23/2026	State Facilities Tax 2%	1.20
08/24/2026	Room Charge	60.00
08/24/2026	Room Hotel Operations Tax 6%	3.60
08/24/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/24/2026	Cook County Hotel Tax 1%	0.60
08/24/2026	MPIRA Hotel Tax 2.5%	1.50
08/24/2026	State Facilities Tax 2%	1.20
08/25/2026	Room Charge	60.00
08/25/2026	Room Hotel Operations Tax 6%	3.60
08/25/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/25/2026	Cook County Hotel Tax 1%	0.60
08/25/2026	MPIRA Hotel Tax 2.5%	1.50
08/25/2026	State Facilities Tax 2%	1.20
08/26/2026	Room Charge	60.00
08/26/2026	Room Hotel Operations Tax 6%	3.60
08/26/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/26/2026	Cook County Hotel Tax 1%	0.60
08/26/2026	MPIRA Hotel Tax 2.5%	1.50
08/26/2026	State Facilities Tax 2%	1.20
08/27/2026	Room Charge	60.00
08/27/2026	Room Hotel Operations Tax 6%	3.60
08/27/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/27/2026	Cook County Hotel Tax 1%	0.60
08/27/2026	MPIRA Hotel Tax 2.5%	1.50
08/27/2026	State Facilities Tax 2%	1.20
08/28/2026	Room Charge	60.00
08/28/2026	Room Hotel Operations Tax 6%	3.60
08/28/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/28/2026	Cook County Hotel Tax 1%	0.60
08/28/2026	MPIRA Hotel Tax 2.5%	1.50
08/28/2026	State Facilities Tax 2%	1.20
08/29/2026	Room Charge	60.00
08/29/2026	Room Hotel Operations Tax 6%	3.60
08/29/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/29/2026	Cook County Hotel Tax 1%	0.60
08/29/2026	MPIRA Hotel Tax 2.5%	1.50
08/29/2026	State Facilities Tax 2%	1.20
08/30/2026	Room Charge	60.00
08/30/2026	Room Hotel Operations Tax 6%	3.60
08/30/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/30/2026	Cook County Hotel Tax 1%	0.60
08/30/2026	MPIRA Hotel Tax 2.5%	1.50
08/30/2026	State Facilities Tax 2%	1.20
08/31/2026	Room Charge	60.00
08/31/2026	Room Hotel Operations Tax 6%	3.60
08/31/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
08/31/2026	Cook County Hotel Tax 1%	0.60
08/31/2026	MPIRA Hotel Tax 2.5%	1.50
08/31/2026	State Facilities Tax 2%	1.20
09/01/2026	Room Charge	60.00
09/01/2026	Room Hotel Operations Tax 6%	3.60
09/01/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/01/2026	Cook County Hotel Tax 1%	0.60
09/01/2026	MPIRA Hotel Tax 2.5%	1.50
09/01/2026	State Facilities Tax 2%	1.20
09/02/2026	Room Charge	60.00
09/02/2026	Room Hotel Operations Tax 6%	3.60
09/02/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/02/2026	Cook County Hotel Tax 1%	0.60
09/02/2026	MPIRA Hotel Tax 2.5%	1.50
09/02/2026	State Facilities Tax 2%	1.20
09/03/2026	Room Charge	60.00
09/03/2026	Room Hotel Operations Tax 6%	3.60
09/03/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/03/2026	Cook County Hotel Tax 1%	0.60
09/03/2026	MPIRA Hotel Tax 2.5%	1.50
09/03/2026	State Facilities Tax 2%	1.20
09/04/2026	Room Charge	60.00
09/04/2026	Room Hotel Operations Tax 6%	3.60
09/04/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/04/2026	Cook County Hotel Tax 1%	0.60
09/04/2026	MPIRA Hotel Tax 2.5%	1.50
09/04/2026	State Facilities Tax 2%	1.20
09/05/2026	Room Charge	60.00
09/05/2026	Room Hotel Operations Tax 6%	3.60
09/05/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/05/2026	Cook County Hotel Tax 1%	0.60
09/05/2026	MPIRA Hotel Tax 2.5%	1.50
09/05/2026	State Facilities Tax 2%	1.20
09/06/2026	Room Charge	60.00
09/06/2026	Room Hotel Operations Tax 6%	3.60
09/06/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/06/2026	Cook County Hotel Tax 1%	0.60
09/06/2026	MPIRA Hotel Tax 2.5%	1.50
09/06/2026	State Facilities Tax 2%	1.20
09/07/2026	Room Charge	60.00
09/07/2026	Room Hotel Operations Tax 6%	3.60
09/07/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/07/2026	Cook County Hotel Tax 1%	0.60
09/07/2026	MPIRA Hotel Tax 2.5%	1.50
09/07/2026	State Facilities Tax 2%	1.20
09/08/2026	Room Charge	60.00
09/08/2026	Room Hotel Operations Tax 6%	3.60
09/08/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/08/2026	Cook County Hotel Tax 1%	0.60
09/08/2026	MPIRA Hotel Tax 2.5%	1.50
09/08/2026	State Facilities Tax 2%	1.20
09/09/2026	Room Charge	60.00
09/09/2026	Room Hotel Operations Tax 6%	3.60
09/09/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/09/2026	Cook County Hotel Tax 1%	0.60
09/09/2026	MPIRA Hotel Tax 2.5%	1.50
09/09/2026	State Facilities Tax 2%	1.20
09/10/2026	Room Charge	60.00
09/10/2026	Room Hotel Operations Tax 6%	3.60
09/10/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/10/2026	Cook County Hotel Tax 1%	0.60
09/10/2026	MPIRA Hotel Tax 2.5%	1.50
09/10/2026	State Facilities Tax 2%	1.20
09/11/2026	Room Charge	60.00
09/11/2026	Room Hotel Operations Tax 6%	3.60
09/11/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/11/2026	Cook County Hotel Tax 1%	0.60
09/11/2026	MPIRA Hotel Tax 2.5%	1.50
09/11/2026	State Facilities Tax 2%	1.20
09/12/2026	Room Charge	60.00
09/12/2026	Room Hotel Operations Tax 6%	3.60
09/12/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/12/2026	Cook County Hotel Tax 1%	0.60
09/12/2026	MPIRA Hotel Tax 2.5%	1.50
09/12/2026	State Facilities Tax 2%	1.20
09/13/2026	Room Charge	60.00
09/13/2026	Room Hotel Operations Tax 6%	3.60
09/13/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/13/2026	Cook County Hotel Tax 1%	0.60
09/13/2026	MPIRA Hotel Tax 2.5%	1.50
09/13/2026	State Facilities Tax 2%	1.20
09/14/2026	Room Charge	60.00
09/14/2026	Room Hotel Operations Tax 6%	3.60
09/14/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/14/2026	Cook County Hotel Tax 1%	0.60
09/14/2026	MPIRA Hotel Tax 2.5%	1.50
09/14/2026	State Facilities Tax 2%	1.20
09/15/2026	Room Charge	60.00
09/15/2026	Room Hotel Operations Tax 6%	3.60
09/15/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/15/2026	Cook County Hotel Tax 1%	0.60
09/15/2026	MPIRA Hotel Tax 2.5%	1.50
09/15/2026	State Facilities Tax 2%	1.20
09/16/2026	Room Charge	60.00
09/16/2026	Room Hotel Operations Tax 6%	3.60
09/16/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/16/2026	Cook County Hotel Tax 1%	0.60
09/16/2026	MPIRA Hotel Tax 2.5%	1.50
09/16/2026	State Facilities Tax 2%	1.20
09/17/2026	Room Charge	60.00
09/17/2026	Room Hotel Operations Tax 6%	3.60
09/17/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/17/2026	Cook County Hotel Tax 1%	0.60
09/17/2026	MPIRA Hotel Tax 2.5%	1.50
09/17/2026	State Facilities Tax 2%	1.20
09/18/2026	Room Charge	60.00
09/18/2026	Room Hotel Operations Tax 6%	3.60
09/18/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/18/2026	Cook County Hotel Tax 1%	0.60
09/18/2026	MPIRA Hotel Tax 2.5%	1.50
09/18/2026	State Facilities Tax 2%	1.20
09/19/2026	Room Charge	60.00
09/19/2026	Room Hotel Operations Tax 6%	3.60
09/19/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/19/2026	Cook County Hotel Tax 1%	0.60
09/19/2026	MPIRA Hotel Tax 2.5%	1.50
09/19/2026	State Facilities Tax 2%	1.20
09/20/2026	Room Charge	60.00
09/20/2026	Room Hotel Operations Tax 6%	3.60
09/20/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/20/2026	Cook County Hotel Tax 1%	0.60
09/20/2026	MPIRA Hotel Tax 2.5%	1.50
09/20/2026	State Facilities Tax 2%	1.20
09/21/2026	Room Charge	60.00
09/21/2026	Room Hotel Operations Tax 6%	3.60
09/21/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/21/2026	Cook County Hotel Tax 1%	0.60
09/21/2026	MPIRA Hotel Tax 2.5%	1.50
09/21/2026	State Facilities Tax 2%	1.20
09/22/2026	Room Charge	60.00
09/22/2026	Room Hotel Operations Tax 6%	3.60
09/22/2026	Chicago Hotel Accommodation Tax 4.25%	2.55
09/22/2026	Cook County Hotel Tax	

Distribution of situations by group

The dataset defines 267 situation types: 169 travel and 98 non-travel. Each situation type is counted once in its group.

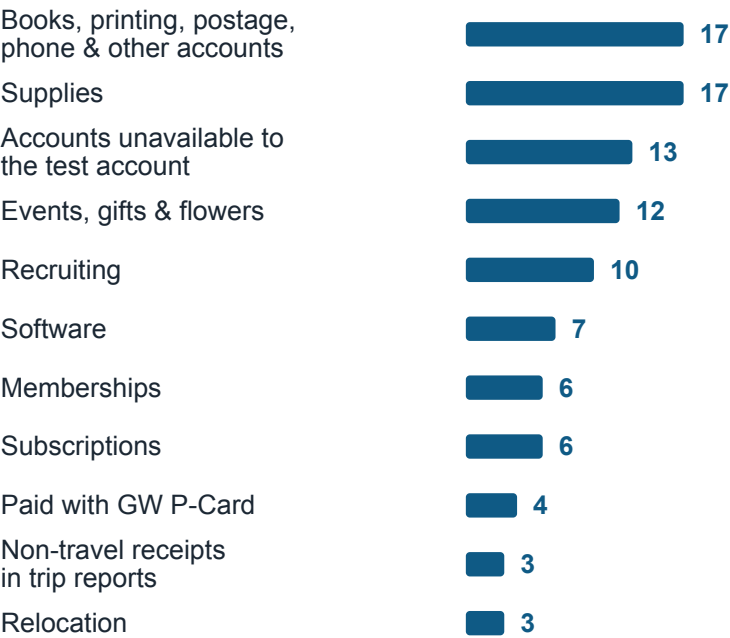
Travel · 169 situations

10 groups · number of situation types



Non-travel · 98 situations

11 groups · number of situation types



Both charts use the same count scale.

Conditions, form work, traps, and the basis of every answer

Each situation also states where it applies, how much form work it takes, whether it hides a deliberate contradiction, and what its answer rests on.

CONDITIONS · WHERE IT APPLIES

city	US city 85 · abroad 18 · local, within 50 miles 9 · any 155 a lunch on a local business day: MEAL-LOCAL-PERSONAL not filed; the same lunch inside an all-day local meeting: MEAL-LOCAL-IN-MEETING file + flag
traveler	staff 22 · student 8 · faculty 13 · anyone 224 an Airbnb stay is flagged for staff (discouraged) and for a student without the advisor's written approval: LOD-AIRBNB-STAFF, LOD-AIRBNB-STUDENT
funding	federal grant 11 · not a grant 8 · any 248 hotel internet: filed on department funds, flagged on a grant (previous pages)

FORM WORK · HOW MUCH FILLING IT TAKES

simple · 204: one line from the receipt, AIR-COACH-DIRECT
complex · 63: more fields or a second screen, such as the folio's four nights, LOD-FOLIO-PERSONAL
Reported apart from the handling, so a score shows whether the agent filled the form badly or judged badly.

Recurrence ⓘ

Different daily amount

📅 Nights: 4

Date	Room Rate *	Room Tax	Tax 2	Tax 3
08/10/2026	297.00	47.53		
08/11/2026	253.00	40.49		
08/12/2026	181.00	28.97		
08/13/2026	235.00	37.61		

(Amounts in USD) [Add Tax Fields](#)

the folio's nights, as the agent entered them in Concur

TRAP · A DELIBERATE CONTRADICTION

19 travel situations hide one, to test whether the agent reads documents against each other.
AIR-WIFI-OTHER-CARRIER file + flag a Wi-Fi receipt from an airline the traveler did not fly: file the fee, flag the mismatch

American Airlines Wi-Fi (Viasat)

inflight wi-fi portal

Purchase Receipt

Aug 24, 2026 6:48 PM Ref: 31068415

Wi-Fi -- full flight pass, AA240610.00

Total\$10.00

Paid: Visa ****4417

Sample document generated for software testing (Expensify/benchmark, benchmark/pod), NOT a real transaction, receipt or invoice.

a Wi-Fi pass on an airline other than the one the trip used

BASIS · WHAT THE ANSWER RESTS ON (SITUATIONS WITH EACH KIND)

- 228 a page of the manual or a GW rule · AIR-COMPANION: manual p.25, a companion only with the VP's approval
- 104 our decision, dated, where the policy is silent · LOD-AIRBNB-STUDENT
- 73 a source we looked up · DUES-SOCIETY-ANNUAL
- 64 experience, said as such: several answers are accepted · GT-TAXI
- 19 one of our own receipts · DOC-PROGRAM

Instance: generated documents with their answer key

An instance is one generated copy of a situation: one to three documents that look real, and their answer key, written by the same code.

THREE INSTANCES OF LOD-FOLIO-PERSONAL · EVERY SITUATION HAS THREE, 801 IN ALL

The Westin Copley Place, Boston

10 Huntington Ave, Boston, MA 02116 ·

Guest Folio

Guest: Changhong Zhang	Room: 620
Arrival: 08/24/2026	Departure: 08/28/2026

Boston
Westin Copley Place
4 nights from 24 August

Palmer House, a Hilton Hotel

17 E Monroe St, Chicago, IL 60603 ·

Guest Folio

Guest: Changhong Zhang	Room: 1436
Arrival: 07/13/2026	Departure: 07/16/2026

Chicago
Palmer House
3 nights from 13 July

Hyatt Regency Atlanta

265 Peachtree St NE, Atlanta, GA 30303 ·

Guest Folio

Guest: Changhong Zhang	Room: 1977
Arrival: 09/14/2026	Departure: 09/19/2026

Atlanta
Hyatt Regency
5 nights from 14 September

What changes: the city, hotel, dates, nights, rates, taxes and the personal charges. **What stays:** the rule tested and the correct handling.

ANSWER KEY · COMPLETE HOTEL EXPENSE RECORD, TRIP A

```
{
  "receipt": "Hyatt Regency Chicago folio.pdf",
  "expense_type": "53104-US LODGING EXPENSE",
  "vendor": "Hyatt Regency Chicago",
  "date": "2026-08-14",
  "amount": "1120.60",
  "currency": "USD",
  "requires": {
    "itemization": {
      "type": "lodging_grid",
      "nights": [
        {
          "date": "2026-08-10",
          "room_rate": "297.00",
          "room_tax": "47.53",
        },
        {
          "date": "2026-08-11",
          "room_rate": "253.00",
          "room_tax": "40.49",
        },
        {
          "date": "2026-08-12",
          "room_rate": "181.00",
          "room_tax": "28.97",
        },
        {
          "date": "2026-08-13",
          "room_rate": "235.00",
          "room_tax": "37.61",
        }
      ]
    }
  },
  "city": "Chicago, Illinois",
  "expense_type_contains_any": ["LODGING", "HOTEL"],
  "date_alternatives": ["2026-08-10"],
  "situation": "LOD-FOLIO-PERSONAL",
  "unreduced_total": "1182.59"
}
```

- HOW GENERATION IS KEPT CONSISTENT
- **Amounts:** computed to the cent before the documents are rendered.
 - **Reproducibility:** the same inputs and random seed reproduce the same files.
 - **Labeling:** every PDF identifies itself as a synthetic test document.
- Generated: 801 instances · 903 files.

Trip: a story and its situations, filed as one expense report

A trip puts a traveler’s story and the situations it contains into one folder, which the agent files as one Concur report, a few documents per turn; the answer key stays hidden from the agent.

THE FOLDER · TRIP A, CHICAGO

the agent sees · 26 documents and the notes

- ▶ Hyatt Regency Chicago folio.pdf
- ▶ Hyatt Regency Chicago reservation confirmation.pdf
- The Purple Pig business lunch check.pdf
- American Airlines DCA–ORD receipt.pdf
- American Airlines seat assignment receipt.pdf
- Uber Aug 11 receipt.pdf
- Uber Aug 11 UPDATED receipt.pdf
- Newsstand receipt.pdf
- Golf with the collaborators.pdf
- Conference program.pdf
- ... 16 more · trip_notes.txt

hidden from the agent

- expected.json – the answer key
- MANIFEST.md – the situations in the trip
- turns.json – the instructions, turn by turn

TRIP_NOTES.TXT, AS THE AGENT READS IT

Trip notes – MOSS 2026 Chicago – Changhong Zhang (postdoctoral researcher, GW School of Engineering)

Purpose: Present a paper at MOSS 2026 (Midwest Optimization & Systems Symposium), Chicago IL, Aug 11–13 2026, and meet the symposium's benchmark working group.

Dates: Aug 10 – Aug 14, 2026 in Chicago.

Filed on: Aug 19, 2026.

Funding: Non-Grant – departmental funds; the department will supply the Oracle Alias, use a placeholder for now.

Remarks and attendees:

- I moved my return flight on Aug 14 from the morning flight to the evening flight (fare difference \$108.00) for a follow-up meeting with the organizing committee that morning; my supervisor approved.
- Aug 12 lunch, The Purple Pig: Lena Hoffmann (University of Toronto), Ana Ferreira (McGill University), me. Benchmark paper planning.

The notes decide several documents: who sat at the Purple Pig lunch, why the return flight was changed.

THE TURNS · ONE RECORDED RUN EACH

turn 1 · 6 files: the flights, the seat, the bag, the changed return, the registration and the tutorial

turn 2 · 2 files: the hotel folio and the reservation confirmation

turn 3 · 7 files: the meals

turn 4 · 11 files: ground transport and the rest

Every turn begins: never submit; read trip_notes.txt first; file what the policy allows, leave out what it does not and say why, split what must be split, and flag every judgment call with its basis.

A TRIP MUST BE CONSISTENT

one way to travel and one ticket for the trip · one hotel at a time · fuel dated the day the car is returned · no two situations at one merchant on one day

23 TRIPS

A–O, 15 travel trips, use all 169 travel situations · P–W, 8 reports outside travel or mixed, use 94 of the 98 others · 382 documents, 257 lines, 134 documents not filed, 192 flags

PER DIEM

on a foreign leg, meals are paid as a daily allowance, so a meal receipt from that leg is not filed (MEAL–INTL–PERDIEM); trip E, New York and London, claims 6 days

Scoring a filed report against its answer key

After a run we read the report back from Concur and check it against the key: trip A passed 84 of 87 checks (96.6%), its folio line passed every one, and the misses say exactly what went wrong.

WHAT THE AGENT FILED · TURN 2

Hyatt Regency Chicago

Chicago, Illinois

✕

Payment Type *

Out of Pocket

Amount *

1,120.60

Currency *

US, Dollar (USD)

✕

Comment

176/500

Folio \$1182.59 reduced by \$61.99: In-Room Movie \$14.99 (8/11), Minibar \$22.00 (8/12), Health Club \$25.00 (8/13) excluded as personal items. Conference rate, 4 nights Aug 10-14.

Nights: 4

Date	Room Rate *	Room Tax	Tax 2	Tax 3
08/10/2026	297.00	47.53		
08/11/2026	253.00	40.49		
08/12/2026	181.00	28.97		
08/13/2026	235.00	37.61		

(Amounts in USD)

Add Tax Fields

Concur after turn 2: the lodging line of 1120.60 with the deduction explained in its comment, and the four nights itemized

SCORE.TXT · TRIP A, BATCH 20260914-02

— per-expense —

...
✓ American Airlines DCA-ORD receipt.pdf 501.40 USD
✓ Hyatt Regency Chicago folio.pdf 1120.60 USD
✗ RPM Steak lunch check (large tip).pdf 70.61 USD MISSING
✗ Uber Aug 11 955 PM.pdf 20.10 USD MISSING

— dimensions —

coverage 21/23
expense_type 21/21
basic_fields 21/21
itemization 1/1
attendees 1/1
skip_respected 5/5
exclusions 1/1
flags_required 5/6
comments_in_concur 6/6
summary_mentions 2/2

△ FLAG MISSING: AIR-SEAT-ECONOMY – expected a flag mentioning one of ['...
△ 1 line(s) in the report match no expected line: 71.77 53102-US BUSINE...
OVERALL: 96.6% | submittable: True

THE TERMS

run: one turn, recorded step by step: the page seen, the decision, the actions, a screenshot. Trip A took 4 turns, 212 steps, \$7.22 and 24 minutes.

export: the report read back from Concur after the run.

score: checks passed ÷ checks in the key, reported by dimension.

not scored by machine: the wording of the business purpose, the Oracle alias, the mileage calculator's fields, whether an approval is attached; a person reads them in the trajectory.

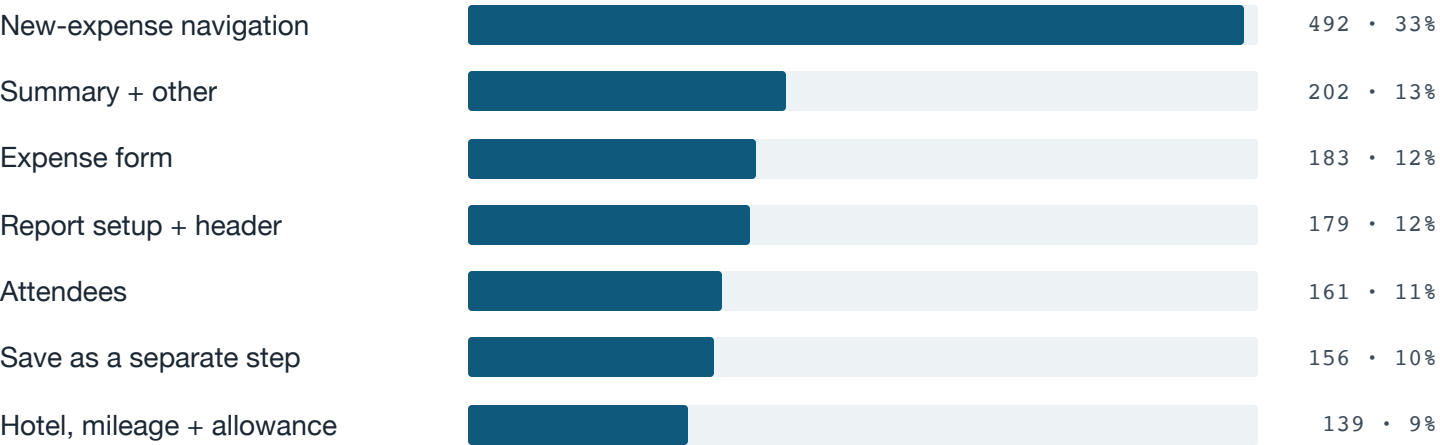
ledger: each situation's record across batches: LOD-FOLIO-
PERSONAL passed 3/3.

For learning across reports this gives, per report, a success score, the steps, the dollars and the seconds; and the same situations recur from trip to trip.

Where the trajectory's steps go

Opening a new expense is the largest phase: three navigation calls recur before the form can be filled. The sample averages 101 model calls per report.

STEPS BY WORKFLOW PHASE · COUNT AND SHARE



Bars start at zero and use the same 0–500 step scale. Shares are rounded.
“Summary + other” includes turn summaries, waits, flags, review, receipts and dismissing dialogs.

WHAT THE SAMPLE MEASURES

A step is one model call, which can issue several actions. A turn is one user instruction; a report can take several turns.

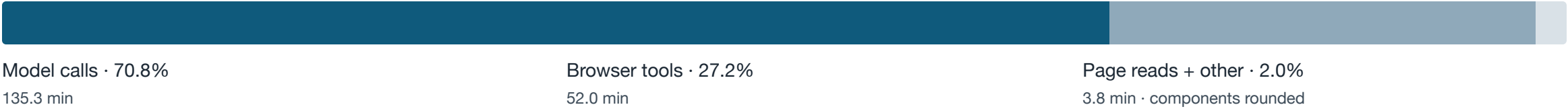
15 trip-runs · 49 turns
1,512 steps · 157 expense lines
100.8 steps per report
9.6 steps per expense line (calc.)

Historical cohort v2: batches 20260914-06 through 20260921-01, after the 14 September interaction fixes. Valid runs with scores; final compliance handoffs and trip R’s known 135-step faulty turn are excluded.

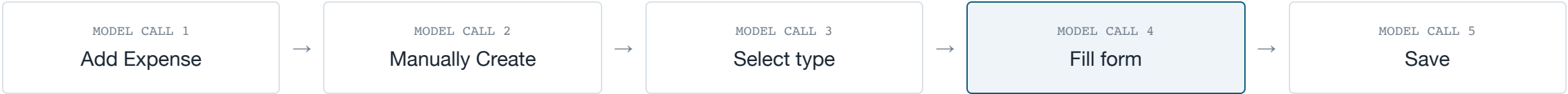
Model calls take most of the running time

Repeated calls are a useful target for acceleration. In the same 15 trip-runs, waiting for the model accounts for 70.8% of recorded elapsed time.

191.2 MINUTES IN TOTAL · 12.7 MINUTES PER REPORT · SUMMED RUNS, EXCLUDING GAPS BETWEEN TURNS



A COMMON ORDINARY EXPENSE LINE · FIVE CALLS



This phase sequence appears in 72 of 126 ordinary expense lines. These lines exclude attendee work, hotel itemization and mileage.

WHAT THE PHASE ANALYSIS SHOWS

Routine workflow phases account for many calls.

They contain 57% of steps and 47.7% of model time: navigation, separate saves, header edits, waits, flags, review and dialog dismissal.

WHAT TO TEST NEXT

Combine predictable actions, with checks at each boundary.

For example, open the expense form as one operation. Phase labels identify candidates; they do not show that every call is removable or establish a measured speedup.

When success is misread, the agent repeats work

Fewer calls require reliable feedback about what changed. The trajectories expose a second problem: successful actions can look like failures and trigger unnecessary retries.

CASE 1 · TRIP R, 21 SEPTEMBER

A saved line is treated as unsaved

Save succeeds → Concur says “Not Found”
→ agent retries saving → keeps retrying

The first \$395 expense was already saved. Misreading the result added 124 steps and \$4.20; the agent eventually asked the user to add it manually, risking a duplicate.

This entire 135-step turn is excluded from the preceding averages. It illustrates a failure loop, not normal report cost.

Implemented: check for the saved line and guard retries.

Still needed: a live comparison to measure the effect.

CASE 2 · HISTORICAL DROPDOWN FEEDBACK

A selected value is reported as unmatched

Option is selected → “No option matched”
→ agent reselects or changes the value

In 20 of 93 such feedback messages, the next observation already showed the selected value. Currency selection could oscillate between USD and GBP.

These counts cover the historical analysis sample, not only v2. They are feedback-message counts, not a v2 action failure rate.

Patched on 24 September: distinguish a failed match from an unconfirmed selection; read the value back.

Still needed: TestTrack and live validation.

The two priorities are complementary: combine repeated navigation, and verify the result before retrying. Evaluate both by steps, elapsed time and the existing report score.

D · Evidence behind the five difficulties (1/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
WebCoach (Liu et al., 2025)	ICLR 2026 MemAgents workshop	External trajectory summary, additional Coach injects suggestions to action model	WebVoyager 643 real-time web tasks; dynamic library accumulates same-stream same-task but excludes same-task-ID retrieval	Skywork-38B + Qwen3-8B Coach: 0.614 vs 0.473; Qwen-VL-7B main table 0.311 vs 0.328	Skywork average per question 215 → 395 s, actions 10.7 → 10.2; time includes extra inference; no token cost measurement	● ○ ●
ReasoningBank (Ouyang et al., 2026)	ICLR 2026	Text experience pool extracted from success and failure trajectories	WebArena, SWE-bench; online single-task stream	WebArena Gemini-2.5-flash success rate 40.5 → 48.8; Shopping judge accuracy 72.7%	WebArena average ReAct steps 9.7 → 8.3; another table total tokens 50,847.4 → 53,054.5, baseline foundation not specified; time and cost not measured	● ○ ●
GenericAgent (Liang et al., 2026)	arXiv	SOP, script, hierarchical memory; fixed tool interface	Multiple models; local task, web, repeated download and research tasks	Same GPT-5.4 repeated download 102 seconds → ~66 seconds; not held-out; five long tasks success 100%, vs Claude Code 100%, OpenClaw 80%	Repeated download tokens 200,439 → 100,000; five long tasks 220.8 s vs OpenClaw 183.1 s; evolution cost not separated	● ● ●
EchoPath (Zhao et al., 2026)	arXiv	Parameterized GUI trajectory memory, precondition and lifecycle	OSWorld-Verified 159 task with established memory; same-task second try change resolution, codex-gpt-5.5-medium	91.2% vs Synapse 91.8%	Median tokens 20,370 vs 586,386; time 127.5 s vs 315.7 s; first build median ~572k tokens/4.5 min counted success-only	● ● ○
MobileGPT (Lee et al., 2024)	ACM MobiCom 2024	Per-app page-subtask graph and action sequence	Mobile apps; cold start then warm start same task with parameter change	GPT-4-turbo; with human repair ablation warm start 98.75%; main table without human repair cannot mix	Human-repair path warm start vs same prompt Derive-only: latency -62.5%, cost -68.8%; offline explore per app 10 to 15 minutes, total \$10.78	● ● ●
ActionEngine (Zhong et al., 2026)	arXiv	State-action graph, locator and descriptor, online general-purpose repair write-back	WebArena 655 tasks, Claude Opus 4.6; template warm-up then eval, patcher off	91.2%; no warm-up graph 73.1%	27 s vs 87 s; \$0.05 vs \$0.40, both per-task relative Claude Code; no explore/warm-up; break-even 39- 101 tasks and missing warm-up	● ● ●
Are Online Skill and Memory Modules Always Worth Their Tokens? (Hajimiri et al., 2026)	EMNLP 2026 main conference (listed in the programme)	Control AWM, ASI, ReasoningBank vs plain executor budget config	WebArena four domain, Gemini 3 Flash, three run	Vanilla-IB success 44.78%, 73.6K total tokens/question, that model four-config best simultaneously	Token only; plain 15 step+pruning, enhance 10 step, near-equal budget	● ● ●
ClawTrace / CostCraft (Yuan et al., 2026)	ACM CAIS 2026 Agent Skills workshop, non- archival	Trajectory-summary exported skill document, frozen model	SpreadsheetBench 30 held-out, OpenClaw + gpt-5.4; also migrate SkillsBench	Remove delete patch after two-type sum degrade 5 → 14; migrate post cost median not decrease	Execution cost with cache; SkillsBench median \$0.143 vs \$0.144; extract cost not reported	● ○ ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.).

D · Evidence behind the five difficulties (2/2)

WORK	VENUE	WHAT IT CHANGES	WHERE IT WAS TESTED	EFFECT, WITH CONDITIONS	TIME, MONEY, LEARNING COST	MARKS
SEDM (Xu et al., 2025)	NeurIPS 2025 SEA workshop	Injectable memory fragments with utility weight and source	LoCoMo, FEVER, HotpotQA; independent build vs evaluation not clear	GPT-4o-mini FEVER: 66 vs G-Memory 62, no memory 57; HotpotQA 39 / 38 / 34	FEVER prompt tokens 2.47M vs G-Memory 3.62M, no memory 1.65M; replay cost and time cost not measured	● ○ ●
HarnessFix (Chen et al., 2026)	arXiv	Prompts, tool description, orchestration, config, adapter and validation script	GAIA, SWE-bench Verified, AppWorld, Terminal-Bench 2.0 Verified; GPT-5 mini	Initial system to patched completion +6.3 to +18.4 percentage points; mean of three runs	AppWorld offline patch 37.2 million tokens; execution-phase time/cost not reported	○ ● ●
ESPO (L. Liu et al., 2026)	EMNLP 2026 Findings (listed in the programme)	Natural-language module prompt	Seven text benchmark, each 500 held-out; Claude Sonnet 4.5	Main optimize single average 74.67% vs GEPA 70.91%; prompt 1,004 vs 1,878 character	PUPA optimize ~3.4M tokens, 2h 13m; no deployment acceleration measured	○ ● ●
GEPA (Agrawal et al., 2026)	ICLR 2026	Module prompts only, model frozen	QA, instructions, privacy rewriting, math; train validation test separated	Qwen3 8B AIME-2025: 32.00 vs MIPROv2 20.00, GRPO 38.00; overall advantage ≠ every task wins	Per-task optimization rollout total 6871 / 3593 / 7051 / 2426 / 1839 / 1839; GRPO per item 24000; not model calls, time or cost	○ ● ●
SICA (Robeyns et al., 2025)	ICLR 2025 Self-Improving Foundation Models Without Human Supervision workshop	Self-complete Python code, tools and oversight mechanism	Sonnet3.5(v2), o3-mini; fixed coding subset and synthetic-file task	SWE peak 17% → 53%, final accuracy 0.51; same baseline for agent selection, not held-out	Cross-four-benchmark full-attempt per-question average: final \$1.91 → \$1.70, 130.2 → 114.5 s; tokens 0.24 → 0.30 M; full cycle ~\$7,000	● ● ○
ADAS (Hu et al., 2025)	ICLR 2025 main conference	Fixed small framework forward() program	GPT-4o search, GPT-3.5 execution; ARC, QA reasoning held-out test	DROP F1 79.4 vs 65.8; MGSM 53.4 vs 39.0, each strongest hand-written baseline	One search + eval ~500 USD (ARC), 300 USD (reasoning); deployment per-question not measured	○ ● ●
Group-Evolving Agents / GEA (Weng et al., 2026)	arXiv	Multiple parent agents share trajectories then modify framework code	SWE-bench/Polyglot; Haiku4.5 then Sonnet4.5, o1 reflection	Report SWE 71.0%, Polyglot 88.3%, but latter inconsistent with claimed 50-question held-out denominator; not use as generalization-gain evidence	Full cycle estimate USD 13,000 / USD 1,500; no per-question post-evolution cost change	○ ● ●

Marks in the last column, in order: time or money of runs measured · learning cost measured · tested on unseen tasks; ● yes, ○ partly, ○ no (initial judgment, calc.).

E1 · The 160 provisions (1/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
AIR-COACH	p.9	Coach/economy for domestic and international; premium economy/business need written pre-approval by president/VP/dean attached; OCONUS segments > 5 h and CONUS segments > 5 h allow premium economy/business for faculty and staff	yes	7
AIR-MEDICAL	p.9	Accommodation other than coach for a medical condition goes through EEO/ER; no medical information on the report	no: an EEO letter is an HR document, never in a receipt folder	0
AIR-FEDERAL-ECON	p.10	Federally funded travel must be booked as economy airfare	yes	1
AIR-LOWEST	p.10	Lowest available airfare: direct route, at most two connections, no extra stays, standard bags, usually a restricted coach fare	yes	3
AIR-MILES	p.9	Tickets bought with frequent-flyer miles are not reimbursed; miles may be used to upgrade if the fare paid is the lowest; the cost of a paid upgrade is the traveler's	yes	2
AIR-IBUY-60	p.9	Out-of-pocket tickets bought through iBuy+ Travel / Travel Leaders can be claimed within 60 days of purchase regardless of travel dates	yes	2
AIR-FLY-AMERICA	p.10	Airfare on a US government grant must be on a US carrier (Fly America / Open Skies)	yes	1
AIR-ANCILLARY-OK	p.10	Standard checked bags, in-flight Wi-Fi and advance seat assignment within economy are paid	yes	4
AIR-ANCILLARY-NO	p.10, App. B p.30	Priority boarding, exit row, extra legroom, Economy Plus, window/aisle seat fees are not paid unless pre-approved (premium economy for segments > 5 h)	yes	3
AIR-TRUSTED-TRAVELER	p.10, App. A p.28	Passport, visa, Global Entry and TSA PreCheck fees for employees who travel routinely, subject to departmental approval	yes	2
AIR-INDIRECT	p.10-11	Indirect or interrupted routes are paid at the lower of the actual fare and the direct-route fare, documented with a screenshot at booking	yes	2
AIR-CHANGE	p.11, App. A p.28	Change/cancellation penalties are allowable when documented with a business reason	yes	3
AIR-UNUSED	p.11	Unused tickets and credits belong to the university; refunds must be requested	yes	2
AIR-CHARTER	p.11	Charter aircraft need prior written approval of the President/EVP/CFO/Provost and Risk Management	no: an executive approval process, not a receipt	0
AIR-CLUB	App. B p.30	Airline club membership fees are unallowable	yes	2
AIR-EXCESS-BAG	App. A p.28, App. B p.30	Baggage of a personal nature (golf clubs, skis, bicycles, surfboards) is unallowable	yes	1
AIR-FIRST	App. B p.30	First class is unallowable except as an EEO medical accommodation	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (2/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
MEAL-BUSINESS	p.11-12	A business meal has two or more persons, business discussed; each attendee's name and the business purpose go on the line; group functions need the announcement or invitation list; attendees may not be an approver of the report	yes	10
MEAL-TRAVEL	p.12	Travel meals are ordinary meals away from home on GW business, at actual cost	yes	6
MEAL-PERDIEM-INTL	p.12	Per diem is used only for international travel (State Department rates built into Concur); per diem days need no receipts	yes	3
MEAL-PERDIEM-CONCUR	p.12	The international meal per diem is claimed through Concur's Travel Allowance itinerary (the State Department rates are built in, pre-populated by the system); meals a host or a conference provided are marked there	yes	2
MEAL-ALCOHOL	p.12-13, App. A p.28	Alcohol is separated from the meal and charged to 52611 – Entertainment; never to a federal grant	yes	6
MEAL-GRANT-UNRESTRICTED	p.12	Alcohol and entertainment that cannot be charged to a sponsored award go to an unrestricted account	yes	5
DOC-75	p.12, p.24	Supporting documents are required for every expense over \$75; itemised receipts for meals over \$75, including meals on a hotel bill	yes	17
DOC-LEVEL3	p.13, App. D p.33	Level 3 / rich card data counts as a receipt for goods only when items are listed and the total matches; never for airfare, hotel, rental car or train	yes	3
DOC-AFFIDAVIT	p.13, p.24-25	The Missing Receipt Affidavit (in Concur) or the Acknowledgement form replaces an itemised meal receipt; missing receipts are never accepted for airfare, hotel, trains or cars	yes	2
LOD-STANDARD	p.13	A standard room, cost kept to a minimum; conference block rates booked directly are fine	yes	3
LOD-P2P	p.13	Peer-to-peer lodging (Airbnb): faculty/staff strongly discouraged; students prohibited without prior written approval of the advisor's office and Risk Management	yes	2
LOD-DEPOSIT	p.14	Deposits should be on a P-Card; deposit receipts list room nights and the rate per night; a group deposit needs the traveler names	yes	1
LOD-GUEST	p.14	Booking a hotel for a visitor: the hotel is paid in full in advance only when the rate requires prepayment; card numbers are never e-mailed	no: arranging a guest's booking is not a traveler's reimbursement	1
LOD-PREPAID	p.14	Prepaid bargain rates are risky: non-refundable and unchangeable	yes	1
LOD-ITEMIZE	p.12-13 (Documentation) + Concur	The folio is itemised by night (room rate + tax per night); personal charges on the folio are left out	yes	4

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (3/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
LOD-NOSHOW	App. B p.30	Hotel cancellation / no-show penalties from failing to cancel a guaranteed reservation are unallowable	yes	2
LOD-LAUNDRY	App. A p.28, App. B p.30	Laundry/dry cleaning only on trips longer than five calendar days	yes	2
LOD-PERSONAL-DAYS	p.25	Non-business days added for personal reasons are not paid unless the stay-over lowers the total cost, with documentation	yes	1
GT-MODE	p.14	Over 200 miles commercial air is generally most economical; 200 miles or less by bus, shuttle, ride service, taxi, rental or private car; more than 200 miles by car needs a quote showing it was most economical	yes	3
GT-AIRPORT	p.15, App. A p.29	Most economical means to and from the airport; a second party driving the traveler is paid at the mileage rate but not more than a taxi or shuttle would cost	yes	7
GT-RENTAL-PREFERRED	p.15	Hertz, Enterprise and National are preferred (insurance included at no extra charge); other companies: staff MUST buy CDW and liability coverage; students must rent from the preferred companies	yes	4
GT-RENTAL-FUEL	p.15, App. A p.28	Decline prepaid fuel and refuel before returning; gasoline for a rental is allowable	yes	4
GT-RENTAL-INTL	p.16	International rentals: staff must buy CDW and liability; students are not authorised to rent abroad	yes	2
GT-RENTAL-CLASS	p.16	Up to full-size; specialty vehicles (luxury, SUV, convertible) are not reimbursed without prior written VP/dean approval; the traveler pays the daily excess	yes	2
GT-RENTAL-STUDENT	p.15	Students: authorised-driver program, 20+, licensed, no 15-passenger vans, not for personal use	yes	3
GT-RENTAL-ACCIDENT	p.16-17	Accidents are reported to the agency, Risk Management and the police	no: an incident report, not an expense	0
GT-MILEAGE	p.16-17, App. A p.28	Private car: mileage at the IRS rate from the normal place of business (not the residence), plus tolls and reasonable parking; commuting is never paid; Concur has a mileage calculator	yes	7
GT-MILEAGE-TWO	p.17	Two or more people in one car: mileage is paid to one person only	yes	1
GT-RAIL	p.17, App. A p.28	Rail and bus at the lowest fare with reserved seating when less expensive than air; evidence of the airfare comparison attached; Acela Business Class is allowable when more economical	yes	4
GT-CAR-IN-LIEU	p.17-18	A car in lieu of air is paid per mileage/rental rules but never more than the lowest available airfare, documented within two business days of approval	yes	2

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (4/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
GT-PRIVATE-CAR	p.18, App. B p.30	Private car services need the Vice President's written pre-approval attached; limos and cabs for extended distances are unallowable	yes	2
GT-CHARTER-BUS	p.18	Charter buses go through Travel Services with contract review	no: a contract, not a receipt	0
GT-FINES	App. B p.30	Fines, tickets, car repairs and car washes are unallowable	yes	3
GT-GAS-PERSONAL	App. B p.30	Gasoline for a personal vehicle is unallowable (mileage covers it)	yes	2
MISC-75	p.18	Miscellaneous expenses over \$75 per transaction must be documented	yes	11
ADVANCE	p.18-19	Travel advances \$200-\$2,500, reconciled within 30 days, unused repaid within 60 days	no: an advance is reconciled against the report, not filed from a receipt	0
ENT-CELEBRATION	p.19, App. B p.30	Cakes, flowers, balloons for employee celebrations are not business expenses; decorations for university events are; gifts for an individual need prior written VP/dean approval attached	yes	10
ENT-TASTE	p.19	Entertainment must advance a university business relationship, within good taste and reasonable limits	yes	3
TIME-30-60	p.6, p.22	Submit within 30 days after the trip; out-of-pocket claims older than 60 days after the trip are not reimbursed; students are exempt from the 60-day limit	yes	5
TIME-PRE-TRIP	p.22	Out-of-pocket travel cannot be claimed before the trip unless booked through iBuy+ Travel	yes	2
REPORT-ONE	p.22	One report per trip; several one-line reports are discouraged	yes	7
PURPOSE	p.22, App. C p.32	The business purpose says what, who, why and the benefit to GW; 'Attend conference' is inadequate	yes	2
VENDOR-INVOICE	p.23, App. B p.30	'Is this a vendor invoice?' is yes only for a true invoice; expenses paid to service providers go through P-Card / invoice processing, and payments to individuals through Payables	yes	17
APPROVAL	p.7, p.23	Nobody approves their own expenses; sponsored expenses route to OVPR	no: routing happens after submission	1
STUDENT-DELEGATE	p.26	A GW employee acting as delegate submits a student's report	no: who submits is a Concur role, not a receipt	0
LOCAL-50	p.25, App. B p.30	Local travel is under 50 miles one way: no personal meals or lodging — unless the mealtime falls within the hours of the meeting	yes	3
COMPANION	p.25	Spouse/companion expenses only with VP pre-approval attached	yes	4
OUTSIDE-SOURCE	App. B p.30	Expenses paid or reimbursed by an outside source are unallowable	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (5/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
FOREIGN–CURRENCY	p.21, p.25	Foreign receipts are recorded in the receipt's currency; Concur converts; never convert by hand	yes	8
PCARD	p.8	The GW P-Card is the preferred payment method; P-Card charges are reconciled in Concur as card transactions, not as out-of-pocket lines	yes	6
APPA–MISC	App. A p.28	Room service, luggage storage fees, valet parking are allowable	yes	6
APPA–TELECOM	App. A p.29	Telephone, fax, internet and data costs are allowable but may not be charged to a federal research project	yes	6
APPA–TIPS	App. A p.29, App. B p.30	Tips are allowable when customary and not excessive; excessive tipping is not	yes	2
APPA–VACCINE	App. A p.29	Vaccinations and other medical costs when required for travel to specific destinations	yes	3
APPA–NONEMLOYEE	App. A p.28	Non-employee travel (applicants, guest lecturers, consultants) paid by GW on a P-Card	no: paid by GW, not reimbursed to the traveler	5
APPB–PERSONAL	App. B p.30–31	Clothing, shoes, jewelry, toiletries, medicines, in-room movies and alcohol, health club, babysitting, recreation, hairdressers, haircuts, personal care	yes	14
APPB–CARDFEES	App. B p.30	Credit-card fees, annual membership, interest and late charges on personal cards	yes	1
APPB–INSURANCE	App. B p.30	Life, accident, baggage, personal property or flight-interruption insurance premiums	yes	1
APPB–LUGGAGE	App. B p.30	Luggage or briefcase purchase or repair	yes	1
APPB–LOST	App. B p.30	Lost, broken or stolen property	yes	1
APPB–READING	App. B p.31	Newspapers, magazines, books for personal use	yes	4
APPB–CHARITY	App. B p.30	Charitable contributions go through University Payables	yes	1
APPB–DEPENDENT	App. B p.30–31	Dependent care, pet care, home or lawn maintenance, other personal costs	yes	3
APPB–CLUBS	App. B p.30	Golf/tennis/health/country club fees, memberships and sporting equipment unless part of a pre-authorised group event	yes	6
APPB–PENALTIES	App. B p.31	Penalties and fees for violating airline, rail, hotel or reservation rules	yes	2
APPB–SPECIAL–OCCASION	App. B p.31	Flowers, fruit, candy, balloons for specific individuals unless approved in advance by a VP or dean, with the approval attached	yes	4
NC–SCOPE	App. A p.28	The manual applies to all expenses incurred for a university purpose regardless of the account charged: a non-travel business expense is judged by the same manual	yes	5

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (6/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC–SELF–PAID	owner decision 2026-09-17 (no. 4); Contract Process Guide (Procurement, 2025-04-30) p.7–8	Subscriptions, office supplies, cloud services and membership dues paid out of pocket ARE reimbursable: file the line, with a notice that GW's preferred channel is P-Card / iBuy+ and that out-of-pocket reimbursement is to be confirmed by the department (GW's order of purchasing methods does not list self-payment, and no GW page allows or forbids it)	yes	19
NC–MIXED–REPORT	owner decision 2026-09-17 (no. 3); https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	Mixed reports are allowed: a non-travel receipt found in a trip folder is filed ON that report with its dedicated expense type, plus a notice that it is a non-travel expense on a travel report (GW: a mix of Domestic and No Travel may pick either travel type)	yes	3
NC–NO–TRAVEL–REPORT	owner decision 2026-09-17 (no. 3); https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	A purely non-travel folder is one report whose header answers 'What type of travel expenses are you claiming?' with No Travel; no Travel Allowance itinerary	yes	39
NC–NOT–IN–CONCUR	owner decision 2026-09-17 (no. 13)	A document that is not reimbursed through a Concur expense report, or for which this account has no expense type, gets no line: the agent says 'not filed in Concur' and names the right channel (Concur Invoice non-employee reimbursement, University Payables / supplier invoice, payroll through the Provost's office, the Gift Card Order Form in Request, GW IT purchasing)	yes	26
NC–TIME–60–TRANSACTION	p.22 (new manual p.23)	Out-of-pocket claims older than 60 days from the transaction date, or from the completion of travel, as applicable, are not reimbursed; a non-travel purchase counts from its transaction date (our reading of 'as applicable'); students and guests are exempt	yes	2
NC–SALES–TAX	p.25 (new manual p.26)	GW's sales-tax exemption does not apply to purchases made with personal funds: the tax on a self-paid receipt is part of the expense (the new manual adds that P-Card holders are responsible for obtaining the exemption)	yes	14
NC–GRANT–ADMIN–COSTS	https://compliance.gwu.edu/direct-charging-administrative-costs-associated-sponsored-projects	Subscriptions, computers, general office supplies, memberships and postage are administrative costs: charged directly to a federal award only when integral to the project, allocable to it, budgeted or approved by the sponsor, and not also recovered as indirect costs	yes	4
NC–CLICK–THROUGH	Contract Process Guide (Procurement, 2025-04-30) p.33–35	Accepting click-through terms is signing a contract; faculty and staff have no authority to sign for the university without a written delegation	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (7/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-OVPR-ROUTING	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	On sponsored funds a report with a type on the OVPR list (52221, 52193, 52441, 52691, 54541, 52641, 53131, 55151, 55541 ...) routes to OVPR; 52211, 54111, 55567, 52199 and 52231 are not on the list	no: routing happens after submission	1
NC-SUBS-BUSINESS	App. B p.31; https://procurement.gwu.edu/appendix-procurement-competitive-exemptions-list (item 13)	Only reading matter FOR PERSONAL USE is unallowable: a journal, database or news subscription for research or teaching is a business expense, type 52221-SUBSCRIPTIONS/PERIODICALS	yes	7
NC-GW-ALREADY-PROVIDES	https://engineering.gwu.edu/software-licenses ; https://research.gwu.edu/aaas-elemental-memberships ; https://gwtoday.gwu.edu/students-leadership-collaborate-expand-free-digital-news-resources ; experience	What GW already provides at no cost — site-licensed software (MATLAB, Microsoft Office), the AAAS Elemental membership, the digital New York Times / Washington Post / Wall Street Journal — is filed when bought separately for business, with a needs_review flag asking why (no GW page forbids the purchase; the handling is our judgment)	yes	1
NC-SW-TYPE-UNNAMED	owner decision 2026-09-17 (no. 5); owner decision D-043 (2026-09-18); https://controller.gwu.edu/capitalized-asset-management	GW names no expense type for software subscriptions and SaaS licences: where the kind is unclear, 52199, 55567, 52441 or 52221 is accepted, with a needs_review flag 'GW does not say which expense type software takes' (the portal calls 52199 'MFA USE ONLY'; the Controller lists it as the general non-capitalized software account). An item that clearly is one kind takes that type and needs no type flag: a usage-billed cloud bill is 55567 (NC-CLOUD-BILL), a journal, database or periodical subscription 52221 (NC-SUBS-BUSINESS)	yes	3
NC-SW-SECURITY-ASSESSMENT	https://procurement.gwu.edu/procurement-methods ; Contract Process Guide (Procurement, 2025-04-30) p.6–7; owner decision 2026-09-17 (no. 6)	All software and SaaS purchases, regardless of dollar amount, must have a GW IT security assessment, and technology hardware, software or cloud services cannot be bought on a P-Card without one: a self-paid software, cloud or AI-tool line carries a needs_review flag saying so (GW IT's own page is narrower: restricted data, a connection to GW systems)	yes	4
NC-AI-TOOLS-LIST	https://it.gwu.edu/ai-tools-status ; https://it.gwu.edu/ai-guidance-and-best-practices ; owner decision 2026-09-17 (no. 6)	AI-tool subscriptions (ChatGPT, Claude) are reimbursable; they are not on GW IT's list of approved AI tools (Microsoft Copilot, Adobe Firefly, GW Chat, Box AI, Zoom AI, Google Gemini, NotebookLM), so the line carries a needs_review flag	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (8/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-CLOUD-BILL	Concur Add Expense dialog, read 2026-09-15 (55567 is in the Other section); https://it.gwu.edu/research-cloud ; owner decision 2026-09-17 (no. 6)	A cloud bill (AWS, Google Cloud, Azure) paid personally is filed as 55567-CLOUD COMPUTING SERVICES, with the self-paid notice and the security-assessment flag; GW IT's Research Cloud offers researchers AWS resources	yes	2
NC-SUPPLIES-TYPE	Contract Process Guide (Procurement, 2025-04-30) p.8; https://ibuy.gwu.edu/office-supplies-and-furniture ; owner decision D-065 (2026-09-18)	Office supplies are 54111-OFFICE SUPPLIES (the main type only: 52691-MISCELLANEOUS OFFICE EXPENSE is not accepted for plain supplies — it takes an office cost that is not a supply, NC-OFFICE-NOT-SUPPLY). GW's preferred way to buy them is an iBuy+ punch-out supplier paid by P-Card; bought out of pocket they are filed with the self-paid notice	yes	7
NC-AMAZON-PERSONAL	https://ibuy.gwu.edu/amazon-business-account ; experience	GW's Amazon Business account takes the P-Card only; an order on a PERSONAL Amazon account is an ordinary self-paid purchase (our reading): file it with the self-paid notice	yes	1
NC-COMPUTER-GWIT	owner decision 2026-09-17 (no. 7); https://it.gwu.edu/faculty-staff-workstation-initiative-guidelines	A computer bought personally is filed under 52193-NON-CAPITALIZED COMPUTING EQUIPMENT when its unit price is under \$5,000, with a needs_review flag: GW IT requires computers to be ordered through GW IT	yes	1
NC-CAPITALIZED-5000	https://controller.gwu.edu/capitalized-asset-management ; owner decision 2026-09-17 (no. 7)	Equipment or software with a unit price of \$5,000 or more and a useful life over one year is capitalized (58524-CAPITALIZED COMPUTER EQUIPMENT AND SOFTWARE), which this account cannot select: not in Concur — GW IT purchasing	yes	2
NC-DUES-PROFESSIONAL	App. B p.30 (a partial listing, silent on professional dues); https://procurement.gwu.edu/appendix-procurement-competitive-exemptions-list (item 12); owner decision 2026-09-17 (no. 4)	The manual has no clause on professional-society dues — App. B names only airline clubs, personal card fees and country, tennis, health or golf clubs. Dues of a professional society are 52211-MEMBERSHIPS, filed with the self-paid notice	yes	5
NC-DUES-PTFPDF	https://facultyaffairs.gwu.edu/professional-development-fund	Part-time Faculty Professional Development Fund: up to \$100 of a professional organisation's membership is reimbursed (at most three awards and \$1,400 per fiscal year), after the department's and the school's approval and the fund's own, claimed through Concur within 30 days of the activity; only part-time faculty eligible for good-faith consideration	yes	2

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (9/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-RECRUIT-MEAL	owner decision 2026-09-17 (no. 8); p.11-12 (new manual p.12); App. C p.32	A meal with a job candidate follows the business-meal rules — every attendee named, the purpose stated, alcohol split off to 52611; the type is 53102-US BUSINESS MEALS or 52641-RECRUITING EXPENSE	yes	2
NC-RECRUIT-EMPLOYEE-PAID	owner decision 2026-09-17 (no. 8); App. A p.28 (new manual p.29)	App. A names non-employee travel (job applicants, guest lecturers, consultants) paid by GW on a P-Card. An employee who paid a candidate's hotel or airfare out of pocket files it — 53131-TRAVEL-NON-GWU PERSONNEL, the ordinary travel type or 52641 — with a needs_review flag	yes	2
NC-RECRUIT-OWN-TRIP	App. C p.32	An employee's own trip to recruit or to interview is ordinary travel ('Travel to University of Washington to recruit students' is one of the manual's model purposes)	yes	1
NC-RECRUIT-JOB-AD	owner decision 2026-09-17 (no. 8); https://hr.gwu.edu/sites/g/files/zaxdzs5691/files/2024-06/gw-advertising-6.pdf ; https://compliance.gwu.edu/advertising-placement	A job advertisement paid personally is filed under 55151-ADVERTISING or 52641 with a needs_review flag: fee-based postings are department funded, normally paid by P-Card or payment request and coordinated by Faculty Affairs or HR; advertising is generally unallowable on sponsored projects	yes	2
NC-RECRUIT-CANDIDATE-RECEIPTS	https://ibuy.gwu.edu/reimbursements-person-not-employee-or-student ; p.26 (new manual p.27)	A candidate's OWN out-of-pocket receipts are not an employee's expenses: a non-employee is reimbursed through Concur Invoice (invoice type 'Non-employee reimbursement': itemized receipts, a memo of the purpose, the payee's address) — not in Concur Expense	yes	1
NC-RECRUIT-SEARCH-FIRM	App. B p.30; https://procurement.gwu.edu/appendix-procurement-competitive-exemptions-list (item 42); https://hr.gwu.edu/special-circumstances	A search firm's invoice is an expense paid to a service provider: University Payables / supplier invoice — not in Concur Expense	yes	1
NC-RECRUIT-SIGNING-BONUS	https://hr.gwu.edu/managing-compensation	A signing bonus is compensation, not an expense: payroll — not in Concur (that it runs through payroll is our inference; the page does not say how it is paid)	yes	1
NC-FOREIGN-VISITOR	p.21 (new manual p.22)	Inviting and paying foreign visitors: the visa type, the activity and the way of paying are settled with the Tax Department BEFORE the invitation; a payment made for a foreign candidate or visitor is filed with a needs_review flag	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (10/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-RELOC-FACULTY-LUMP-SUM	https://provost.gwu.edu/sites/g/files/zaxdzs5926/files/2025-07/2025%20GW%20Faculty%20Handbook.pdf §2.3; https://facultyaffairs.gwu.edu/relocation-and-moving-allowance-new-faculty ; owner decision 2026-09-17 (no. 9)	A new full-time faculty member's relocation is a lump sum from the Provost's office by rank — \$3,000 assistant professor or instructor, \$4,500 associate or full professor — paid through payroll and taxable, for any expense of the move including house-hunting trips; receipts only prove the move. Not a Concur expense report, and this account has no relocation type (53211-RELOCATION EXPENSE is not selectable)	yes	2
NC-RELOC-PARTTIME-NONE	https://facultyaffairs.gwu.edu/relocation-and-moving-allowance-new-faculty ; owner decision 2026-09-17 (no. 9)	Part-time faculty, and full-time faculty appointed for one semester, have no relocation allowance	yes	1
NC-RELOC-STAFF	https://hr.gwu.edu/why-gw ; owner decision 2026-09-17 (no. 9)	Staff relocation: GW publishes no policy (the HR page lists only third-party moving discounts)	no: no GW rule to test against — the owner's decision is not to invent one	0
NC-EVENT-FLOWERS	p.19 (new manual p.19–20, which adds 'flowers')	Table centerpieces, flowers and decorations for university events are allowable	yes	2
NC-EVENT-TYPE-52612	owner decision 2026-09-17 (no. 11); App. A p.28 (new manual p.29): 'alcohol must be charged to account 52611'	Gifts for an individual and decorations or flowers for a university event are 52612-SPECIAL EVENTS/BUSINESS RELATIONS (the portal asks for attendees on it); 52611-ENTERTAINMENT is alcohol and the entertainment of business contacts	yes	5
NC-GIFT-CARDS	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines ('Purchase of Gift Cards', read 2026-09-18); owner decision 2026-09-17 (no. 13)	Gift cards and cash equivalents need permission BEFORE the purchase: 'you must request permission to make the purchase via a Gift Card Order Form in iBuy+ under the Requests tab'. A card bought personally without it is not claimed on an expense report — not in Concur Expense (that it is then not filed at all is our reading: the page says nothing about claiming it afterwards)	yes	1
NC-PCARD-RECEIPT-FILED	owner decision 2026-09-17 (no. 10); p.8	A receipt that shows it was paid with the GW P-Card is FILED anyway — Payment Type can only be Out of Pocket, this account has no card feed — with a needs_review flag and the line Comment 'P-Card transaction; the card is not linked to this account; once it is linked, replace this line with the card transaction or delete it.' (the original AIR-PCARD-PAID and REG-PCARD follow it too since 2026-09-18, with lesson L-023 rewritten)	yes	4

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (11/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-BOOKS	App. B p.31; owner decision D-019	A book for research or teaching is a business expense (only books for personal use are unallowable): type 52231-BOOKS	yes	1
NC-PRINTING	experience (the manual does not mention printing or copying); Concur Add Expense dialog	Printing, copying and binding by an outside shop: 55511-NON-GW PRINTING/REPRODUCTION, 52661-OFFICE COPYING - NOT GWU or 55513-BINDING	yes	2
NC-POSTAGE	experience (the manual does not mention postage); https://compliance.gwu.edu/direct-charging-administrative-costs-associated-sponsored-projects	Postage and express mail for university business: 53411-POSTAGE/EXPRESS MAIL - NON-GW; an administrative cost on a federal award	yes	1
NC-TELEPHONE-TYPE	App. A p.29; Concur Add Expense dialog; owner decision D-065 (2026-09-18)	Telephone, internet and data costs are allowable but never on a federal research project; outside a trip the type is 53311-TELEPHONE - DIRECT (the main type only: 53313-LONG DISTANCE USAGE is not accepted for a phone, hotspot or data plan)	yes	1
NC-SEMINAR-FEES	App. A p.28 (new manual p.29)	Fees for attendance at seminars and conference registration are allowable: 53241-SEMINAR FEES or 53251-CONFERENCE/SEMINAR/TRAINING EXPENSE	yes	4
NC-TRAINING-COURSE	experience; App. A p.28 (new manual p.29) lists conference and seminar fees only; owner decision D-068 (2026-09-18)	A job-related training course (an online course with a certificate that the supervisor or the project's PI asked for) is filed under 53251-CONFERENCE/SEMINAR/TRAINING EXPENSE, the one type whose name says training, with a needs_review flag: the manual lists conference registration and seminar fees as allowable and says nothing about training courses, so the type is our reading, to be confirmed	yes	1
NC-SUBJECT-PAYMENTS	p.21 (new manual p.22); https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	Funds for distribution to human subjects can be advanced through iBuy+ Expense; payments to research subjects are 55541-PAYMENTS TO SUBJECTS (on the OVPR list). Gift cards for subjects go through the Gift Card Order Form, not a line (NC-GIFT-CARDS); whether a subject payment other than gift cards is filed under 55541-PAYMENTS TO SUBJECTS was not verified — no situation yet	yes	1
NC-OVPR-NOT-SELECTABLE	owner decision 2026-09-17 (no. 13); owner decisions D-048, D-058 (2026-09-17) and D-067 (2026-09-18); https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	An account on GW's OVPR list that this test account cannot select (52131, 52615, 53132, 53211, 55161, 55561, 58323, 58417, 58516, 58522, 58523, 58524, 58535, 58536, 58691) gets no line: the agent says the document is not filed in Concur, names the account by its code and name, and names the channel — never a nearby selectable type in its place	yes	13

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (12/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-TYPE-UNDEFINED	owner decision D-055 (2026-09-17); experience	Where GW publishes no definition of the accounts that could take a line (52691 against 52441, 52721 against 53241 and 53251, 56311 against 55591), the defensible types are all accepted and the line carries a needs_review flag that the type is to be confirmed, in words tied to that line	yes	5
NC-PERMIT-FEE	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines (52441 on the OVPR list); https://www.faa.gov/uas/getting_started/register_drone; experience	A permit, licence or registration fee a lab pays for university business (the FAA registration of research drones) is 52441-PERMITS LICENSES & FEES — GW publishes no definition; not 58335-FEES/PERMITS, a capital-project type in our reading, nor 53111	yes	1
NC-EVENT-FOREIGN	p.19 (new manual p.19–20); owner decision 2026-09-17 (no. 11); p.21, p.25 (new manual p.22, p.26)	Flowers or decorations for a university event bought on a foreign leg are 52613-NON US SPECIAL EVENTS/BUSINESS RELATIONS, the Foreign Travel Expense twin of 52612, recorded in the currency paid	yes	1
NC-RECRUIT-BOOTH	Contract Process Guide (Procurement, 2025-04-30) App. A p.50 (item 36); App. B p.30; owner decision 2026-09-17 (no. 8, by analogy)	An employer booth at a career fair paid personally is filed under 52641-RECRUITING EXPENSE with a needs_review flag: booth space is normally paid by the department on a P-Card or a Payment Request, and App. B puts payments to a service provider under the P-Card and invoice-processing policies	yes	1
NC-OFFICE-NOT-SUPPLY	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines (52691 on the OVPR list); Concur Add Expense dialog (52691 heads the Supplies section); experience	An office cost that is not a supply — a notary fee for a university document — is 52691-MISCELLANEOUS OFFICE EXPENSE; 52441 is accepted too, as GW defines neither (NC-TYPE-UNDEFINED)	yes	1
NC-PROGRAM-DEVELOPMENT	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines; https://compliance.gwu.edu/identification-and-treatment-unallowable-costs-sponsored-projects; App. A p.28 (new manual p.29); Contract Process Guide (Procurement, 2025-04-30) App. A p.48 (item 14)	A fee paid to develop a new academic program (a program-design workshop) is 52721-PROGRAM DEVELOPMENT ACTIVITY — read as not a fundraising account, since GW lists 52722 but not 52721 as unallowable on sponsored projects; 53241 and 53251 are accepted too (NC-TYPE-UNDEFINED)	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (13/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-MISC-SUPPLIES	https://ibuy.gwu.edu/expense-reporting-helpful-guidelines (54541 on the OVPR list, 54111 not); https://ibuy.gwu.edu/facilities-and-mro-supplies ; owner decision D-064 (2026-09-18)	Supplies that are neither office nor laboratory supplies — moving boxes and packing material for a lab move — are 54541-MISCELLANEOUS SUPPLIES, not 54111 or 54121; bought out of pocket they are filed without the preferred-channel notice, which D-064 keeps to subscriptions, cloud services, office supplies and dues	yes	1
NC-CONSULTANT-TRAVEL	App. A p.28 (new manual p.29); Contract Process Guide (Procurement, 2025-04-30) p.3, p.45–46; https://taxdepartment.gwu.edu/worker-classification ; owner decision 2026-09-17 (no. 8, by analogy)	An employee who paid a contracted consultant's airfare files it under 55112-CONSULTANTS TRAVEL (53131 accepted) with a needs_review flag: GW pays a consultant's travel itself on a P-Card or on the consultant's invoice under the agreement; the consultant's own receipts are a non-employee reimbursement in Concur Invoice	yes	1
NC-PROMO-ITEMS	https://business-services.gwu.edu/sites/g/files/zaxdzs6001/files/downloads/LTP_1920_8_Departmental%20Guidelines_Remediated.pdf ; Contract Process Guide (Procurement, 2025-04-30) p.18; owner decision D-016 (2026-09-14); experience	Promotional items bearing GW's marks are 55152-PUBLICITY & PROMOTION, made by a GW-licensed vendor after Licensing & Trademarks approved the artwork: the approval is attached to the line (our judgment), and without it the line is filed with a needs_review flag asking for it	yes	3
NC-PROMO-GRANT	https://compliance.gwu.edu/identification-and-treatment-unallowable-costs-sponsored-projects ; https://www.law.cornell.edu/cfr/text/2/200.421 ; p.12	55152-PUBLICITY & PROMOTION is on GW's list of costs unallowable on sponsored projects and 2 CFR 200.421 bars promotional items: on a federal award the line carries a needs_review flag to move it to an unrestricted account	yes	1
NC-SUBJECT-ADVANCE	p.21 (new manual p.22); https://ibuy.gwu.edu/sites/g/files/zaxdzs4686/files/downloads/EXP%20%20Cash%20Advances%20-%20Requesting%20a%20Cash%20Advance%20032219.pdf ; https://ibuy.gwu.edu/sites/g/files/zaxdzs4686/files/2024-08/EXP%20Cash%20Advances%20-%20Adding%20to%20a%20Report%20and%20Returning%20a%20Balance%20Aug%202024.pdf ; https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	Payments to research subjects made from a human-subjects cash advance are 55541-PAYMENTS TO SUBJECTS, filed once the subjects were paid, with the signed payment log; every advance must be applied to an expense report, and as this test account holds no advance the line carries a needs_review flag	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (14/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-SUBJECT-CASH-PERSONAL	App. B p.31; https://procurement.gwu.edu/sites/g/files/zaxdzs4691/files/2024-05/forte_participants_payment_procedures_052024.pdf; https://procurement.gwu.edu/procure-pay-resources; owner decision 2026-09-17 (no. 13)	Cash a researcher paid study participants from their own pocket is not filed in Concur although 55541 is selectable: GW's route is Advarra (Forte) Participant Payments, or a method the Finance Division approves before the study (that the cash cannot be claimed afterwards is our reading)	yes	1
NC-ROOM-SHORT-TERM	https://procurement.gwu.edu/gw-hotel-event-requests; Contract Process Guide (Procurement, 2025-04-30) p.3, p.33-35; https://ibuy.gwu.edu/expense-reporting-helpful-guidelines	A meeting room rented for part of a day is 55591-BUILDING/ROOM/FACILITY RENTAL (SHORT TERM), not lodging or an event cost; GW sources hotel meeting and event space through Procurement (Hotel & Event Request), so a room an employee booked and paid is filed with a needs_review flag	yes	1
NC-ROOM-OCCUPANCY	Contract Process Guide (Procurement, 2025-04-30) p.4, p.46; https://finance.gwu.edu/departments; owner decision D-055 (2026-09-17)	A monthly desk or space membership is 56311-BUILDING/ROOM/FACILITY RENTAL OCCUPANCY (55591 accepted: GW defines neither); a licence for the use of space is a real-estate transaction for GW's Office of Real Estate that an employee does not sign, so the line carries a needs_review flag for each question	yes	1
NC-CONTRACT-SIGNATURE	Contract Process Guide (Procurement, 2025-04-30) p.33-34; https://procurement.gwu.edu/media/2141 p.34 (§6.1.2)	Only individuals with a specific delegation of authority sign contracts that bind GW for goods, services, equipment or construction: a lease or a contractor's proposal an employee signed may not bind GW and the signer may become personally liable	yes	2
NC-LEASE-COPIER	Contract Process Guide (Procurement, 2025-04-30) p.3, p.46; https://procurement.gwu.edu/multi-function-device-suppliers; https://procurement.gwu.edu/media/2141 p.34 (§6.2)	A lease of office equipment is a contract P2P processes (52131-LEASE-OFFICE EQUIPMENT, which this account cannot select): new copiers come only from GW's two contracted multifunction-device suppliers, requested on the order placement form through Gatekeeper; a payment on a lease an employee signed is not filed in Concur (not 52195 or 52691)	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (15/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-EMPLOYEE-AWARD	https://taxdepartment.gwu.edu/processing-gifts-prizes-and-awards-employees ; https://hr.gwu.edu/recognition-and-positive-vibes ; https://compliance.gwu.edu/taxation-gifts-prizes-and-awards-employees ; p.19, App. B p.30	Gifts, prizes and awards to employees — a recognition outing the chair approved as an On-the-Spot award — are 52615-EMPLOYEE SPECIAL ACTIVITY, which this account cannot select: not filed in Concur; University Payables pays or reimburses non-taxable awards through the Payment Request Form, and a missing expense type is asked of the iBuy Help Desk; not 52611, 52612 or 57121	yes	1
NC-TRAINEE-TRAVEL	https://grants.nih.gov/grants/policy/nihgps/HTML5/section_11/11.3.8_allowable_and_unallowable_costs.htm (11.3.8.5); https://ibuy.gwu.edu/expense-reporting-helpful-guidelines ; https://sponsoredprojects.gwu.edu/award-expenditures ; experience	Travel of a trainee appointed on a training grant is 53132-TRAINEE TRAVEL, which this account cannot select: not filed in Concur — the grant's program coordinator and the Research Support Team charge it to the training grant (our reading: GW publishes no trainee-travel procedure), or the iBuy Help Desk is asked to add the type; no ordinary 531xx type in its place	yes	1
NC-FOREIGN-SERVICE-PAYEE	App. B p.31; https://procurement.gwu.edu/media/2141 p.8; https://procurement.gwu.edu/international-transactions ; Contract Process Guide (Procurement, 2025-04-30) App. A p.50 (item 31); https://taxdepartment.gwu.edu/worker-classification-and-international-engagement-request-form-exemptions	A foreign individual's unpaid invoice for a professional service (a translation) is 55161-NON-US PROFESSIONAL SERVICES, which this account cannot select: University Payables pays it — the payee registers as a supplier with form W-8BEN and the approved invoice is paid by EFT through a Payment Request, no purchase order; services of international individuals are never paid on a P-Card	yes	1
NC-SERVICE-CONTRACT	Contract Process Guide (Procurement, 2025-04-30) p.5, p.34, p.40; https://procurement.gwu.edu/media/2141 p.8; p.23 (new manual p.24)	A service bought under a statement of work is a contract P2P places — a purchasing requisition with the SOW, a purchase order, the invoices to University Payables — account 55561-CONTRACT SERVICE-OTHER in our reading, which this account cannot select; an unsigned statement of work is not proof of payment and nothing is filed	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (16/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-CONSTRUCTION-FPCM	Contract Process Guide (Procurement, 2025-04-30) p.16, p.33-34; https://procurement.gwu.edu/media/2141 p.34; App. B p.30	Construction or renovation work in a GW building is 58323-OTHER CONSTRUCTION CONTRACTS, which this account cannot select: FPCM (Facilities Planning, Construction and Management) manages it and administers the construction contracts with P2P; a deposit an employee paid is not filed (not 58327-FIXED EQUIPMENT)	yes	1
NC-CAPITAL-ASSETS	https://controller.gwu.edu/capitalized-asset-management; https://compliance.gwu.edu/fixed-asset-management; https://compliance.gwu.edu/standardization-goods-equipment-and-services-purchasing; Contract Process Guide (Procurement, 2025-04-30) p.3	Furniture, copiers and laboratory instruments of \$5,000 or more a unit (shipping, handling and installation included) with a useful life over one year are capitalized — 58417, 58535, 58536, none of them selectable: bought through Procurement (a purchase order in EAS or iBuy+), capitalized and tagged by the Controller's Fixed Assets department; not filed in Concur, and no part of the invoice becomes a line	yes	3
NC-VEHICLE-FLEET	Contract Process Guide (Procurement, 2025-04-30) p.3, p.34; https://transportation.gwu.edu/fleet; https://controller.gwu.edu/capitalized-asset-management	A vehicle for GW is 58516-CAPITALIZED VEHICLES EXPENSE, which this account cannot select: P2P handles vehicle leasing and purchasing and Transportation and Logistics administers GW's vehicles (the shared motor pool); a deposit an employee paid on one is not filed	yes	1
NC-FABRICATION	https://compliance.gwu.edu/fixed-asset-management; https://controller.gwu.edu/capitalized-asset-management; https://www.acquisition.gov/far/52.245-1	Parts of a fabrication — a unique instrument of more than \$5,000 assembled from parts, some or all under \$5,000 — are capital even when each part is under \$5,000: 58522-GW OWNED CAPITALIZED FABRICATION EQUIPMENT, or 58523-NON-GW OWNED CAPITALIZED FABRICATION EQUIPMENT when title vests in the sponsor; neither is selectable — the parts are bought through Procurement and the Fixed Assets department records the fabrication	yes	2
NC-GOVERNMENT-PROPERTY	https://www.acquisition.gov/far/52.245-1; https://compliance.gwu.edu/fixed-asset-management; https://sponsoredprojects.gwu.edu/award-expenditures; experience	Equipment whose title vests in the Government under a federal cost-reimbursement contract (FAR 52.245-1) is not a GW asset: 58691-EQUIPMENT-NOT GWU in our reading (no GW page defines it), which this account cannot select — bought through Procurement on the contract and recorded as government property by the Fixed Assets department	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (17/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-SUPPLIES-BY-KIND	Concur Add Expense dialog, Supplies section (EXPENSE_CATEGORIES.md §1.1, read 2026-09-15); owner decision D-065 (2026-09-18); experience	The Supplies section names the KIND of the goods — 54111 office, 54112 library, 54121 laboratory, 54131 linen, 54141 instructional, 54151 printing, 54161 athletic, 54531 photographic, 54541 miscellaneous: supplies of a named kind take that account, not the office type (54111) and not the residual one (54541, which is for supplies of no named kind — NC-MISC-SUPPLIES). Where the kind of the goods is itself arguable the defensible accounts are all accepted, with the type flag (NC-TYPE-UNDEFINED)	yes	5
NC-LAB-SUPPLIES	https://ibuy.gwu.edu/lab-supplies-equipment-0; https://compliance.gwu.edu/direct-charging-administrative-costs-associated-sponsored-projects; https://sponsoredprojects.gwu.edu/budget-cost-categories; Concur Add Expense dialog (54121 in the Supplies section)	Consumables of a laboratory — reagents, gloves, pipette tips, dry ice — are 54121-LABORATORY SUPPLIES. They are a DIRECT cost of the research, not one of the administrative costs GW's direct-charging page lists (subscriptions, computers, general office supplies, memberships, postage): on a federal award a laboratory-supply line needs no administrative-cost flag and no sponsor-approval question (NC-GRANT-ADMIN-COSTS does not reach it)	yes	2
NC-LAB-EQUIPMENT	https://controller.gwu.edu/capitalized-asset-management; Concur Add Expense dialog, Equipment section; owner decision D-079 (2026-09-20)	Furniture and equipment under \$5,000 a unit is not capitalized (NC-CAPITALIZED-5000): the Equipment section splits it in two — 52191-NON-CAPITALIZED INSTRUCTIONAL & LAB FURNITURE EQUIPMENT for what furnishes a laboratory or a classroom, 52195-OTHER NON-CAPITALIZED FURNITURE & EQUIPMENT for the rest of an office. A computer stays 52193 (NC-COMPUTER-GWIT); an office device is equipment, not a supply (54111) and not a miscellaneous office expense (52691)	yes	2
NC-STANDARDIZATION	https://compliance.gwu.edu/standardization-goods-equipment-and-services-purchasing; Contract Process Guide (Procurement, 2025-04-30) p.3, p.8	GW asks departments to buy furniture, equipment and services to the university's standards, through Procurement and its contracted suppliers, and to consult P2P before departing from them: furniture an employee bought out of pocket is filed with a needs_review flag naming the standardization policy (the page does not say the purchase is void — the flag asks, it does not refuse)	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E1 · The 160 provisions (18/18)

PROVISION	PAGE OR SOURCE	THE RULE, AS RECORDED	CAN AN AGENT ACT ON IT?	SITUATIONS
NC-BINDING-ALONE	Concur Add Expense dialog, Printing/Copying section; experience (GW's manual does not mention printing, copying or binding)	An outside shop's invoice for BINDING alone — no printing on it — is 55513-BINDING, the section's own type; 55511-NON-GW PRINTING/REPRODUCTION is printing and reproduction and 52661-OFFICE COPYING - NOT GWU is copying (NC-PRINTING covers an invoice that mixes them)	yes	1
NC-AV-LIBRARY-SERVICES	Concur Add Expense dialog, Books/Subscriptions section; experience	A library's SERVICE — document delivery, a copyright-cleared article, digitisation, an audiovisual service — is 52716-AUDIOVISUAL/LIBRARY SERVICES, next to 52221-SUBSCRIPTIONS/PERIODICALS and 52231-BOOKS in the same section: a service bought item by item is neither a subscription (nothing recurs) nor a book	yes	1
NC-FLEET-FUEL	https://transportation.gwu.edu/fleet; https://transportation.gwu.edu/fleet-rental; App. B p.30; Concur Add Expense dialog (54511 is the Vehicle Expense section's only type)	Fuel an employee paid for a GW FLEET vehicle is filed under 54511-FUEL-GASOLINE AND OIL, with a needs_review flag: Transportation and Logistics settles a fleet rental's final costs 'by an interdepartmental assessment via a journal entry' and publishes nothing about fuel, so whether the fill-up belongs in that charge is to be confirmed. App. B's unallowable fuel is the fuel of a PERSONAL car, which mileage already pays for (GT-GAS-PERSONAL); a rental car's fuel on a trip stays with the trip's ground transportation	yes	1

Travel provisions first (83, from the manual, rev. 2024-02-21), then those for expenses outside travel (77). Situations: how many situations cite the provision. Source: ExpenseAI benchmark/pool/coverage.py and coverage_newcats.py.

E2 · The 267 situations (1/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
AIR-COACH-DIRECT	Coach ticket bought from the airline	airfare	file	simple	domestic	manual, receipt	A P
AIR-COACH-INTL	Coach ticket to a foreign city, bought in USD	airfare	file	simple	foreign	manual, receipt	B H P
AIR-AGENCY	Coach ticket booked through iBuy+ Travel / Travel Leaders	airfare	file	simple	domestic	manual, receipt	E
AIR-IBUY-PRETRIP	iBuy+ ticket for NEXT month's trip, claimed now	airfare	file + flag	simple	domestic	manual	F
AIR-00P-PRETRIP	Airline-direct ticket for a trip not yet taken	airfare	not filed	simple	domestic	manual	K
AIR-BAG	First checked bag fee	airfare	file	simple	any	manual, receipt	A P
AIR-BAG-EXCESS	Sports-equipment baggage fee (bicycle)	airfare	not filed	simple	any	manual	E
AIR-WIFI	In-flight Wi-Fi	airfare	file	simple	any	manual	C
AIR-WIFI-OTHER-CARRIER	In-flight Wi-Fi receipt from an airline the traveler did not fly	airfare	file + flag	complex	any	experience, manual	I
AIR-SEAT-ECONOMY	Paid advance seat assignment inside economy	airfare	file + flag	simple	any	experience, manual	A
AIR-EXTRA-LEGROOM	Extra-legroom seat upgrade	airfare	file + flag	simple	any	manual, decision	F
AIR-PRIORITY	Priority boarding	airfare	file + flag	simple	any	manual, decision	F
AIR-PREMIUM-NOMEMO	First class on a short domestic flight, no approval in the folder	airfare	file + flag	simple	domestic	experience, manual, decision	I
AIR-PREMIUM-LONGHAUL-MEMO	Business class on a 7-hour international segment, with the dean's memo	airfare	file	complex	foreign; staff	manual	G
AIR-PREMIUM-LONGHAUL-NOMEMO	Business class on a 7-hour segment, memo missing	airfare	file + flag	simple	foreign; staff	manual	O
AIR-ECREDIT	Ticket paid partly with an airline eCredit of unknown origin	airfare	file + flag	simple	any	manual, receipt, looked up, decision	C
AIR-MILES-AWARD	Award ticket paid with miles (only taxes charged)	airfare	not filed	simple	domestic	manual	H
AIR-MILES-UPGRADE-COPAY	Coach fare plus a cash co-pay for a miles upgrade	airfare	file part	simple	domestic	manual	F
AIR-REFUNDED	Ticket later refunded in full (refund confirmation in the folder)	airfare	not filed	simple	domestic	manual, decision	B
AIR-CHANGE-NOREASON	Fare difference for a changed flight, no reason given	airfare	file + flag	simple	domestic	manual, looked up	F

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (2/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
AIR-CHANGE-REASON	Fare difference for a changed flight, reason in the notes	airfare	file	simple	domestic	manual	A
AIR-CANCEL-PENALTY	Airline cancellation penalty for a business-caused cancellation	airfare	file	simple	domestic	manual	H
AIR-COMPANION	Ticket for the traveler's spouse on the same booking	airfare	not filed	simple	domestic	manual	D
AIR-COMPANION-MEMO	Spouse's ticket with the Vice President's written pre-approval	airfare	file	complex	domestic	manual	H
AIR-INDIRECT-COMPARISON	Personal stop-over routing, with the direct-fare screenshot	airfare	file part	simple	domestic	manual	I
AIR-INDIRECT-NOCOMPARISON	Personal stop-over routing without a fare comparison	airfare	file + flag	simple	domestic	manual	K
AIR-FOREIGN-CARRIER-GRANT	Non-US carrier on a federal grant (Fly America)	airfare	file + flag	simple	foreign; grant	manual	E
AIR-CLUB	Airport lounge day pass	airfare	not filed	simple	any	manual	E
AIR-INSURANCE	Trip / baggage insurance bought with the ticket	airfare	not filed	simple	any	manual	D
AIR-GLOBAL-ENTRY	Global Entry / TSA PreCheck application fee	airfare	file + flag	simple	any; staff	manual	E
AIR-PASSPORT	Passport renewal / visa fee before an international trip	airfare	file + flag	simple	foreign; staff	manual	B
AIR-PCARD-PAID	Ticket paid with the GW P-Card	airfare	file + flag	simple	domestic	experience, manual, decision	K
AIR-OUTSIDE-REIMBURSED	Ticket the host institution reimburses	airfare	not filed	simple	domestic	manual	O
LOD-FOLIO	Hotel folio, several nights at varying rates	lodging	file	complex	domestic	manual, receipt, looked up	B F P W
LOD-FOLIO-PERSONAL	Folio with a movie, minibar and health-club charge on it	lodging	file part	complex	domestic	manual, receipt	A
LOD-FOLIO-LAUNDRY-SHORT	Laundry on a four-day trip	lodging	file part	complex	domestic; max days=4	manual	M
LOD-FOLIO-LAUNDRY-LONG	Laundry on a seven-day trip	lodging	file	complex	domestic; min days=6	manual	D J
LOD-FOLIO-PARKING	Hotel valet parking on the folio	lodging	file	complex	domestic	manual	N
LOD-FOLIO-ROOMSERVICE-BIG	Room-service dinner over \$75 on the folio, no itemised ticket	lodging	file + flag	complex	domestic	manual	F
LOD-FOLIO-ROOMSERVICE-SMALL	Room-service breakfast under \$75 on the folio	lodging	file	complex	domestic	manual	K

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (3/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
LOD-FOLIO-INTERNET	Hotel internet charge on the folio	lodging	file	complex	domestic	manual	K
LOD-FOLIO-INTERNET-GRANT	Hotel internet on a federal research grant	lodging	file + flag	complex	domestic; grant	manual	C
LOD-FOLIO-INROOM-ALCOHOL	In-room bar alcohol on the folio	lodging	file part	complex	domestic	manual	I
LOD-OTA	Expedia-style prepaid receipt: nightly prices and one tax total	lodging	file	complex	domestic	experience, manual, receipt	E
LOD-FOREIGN	Foreign hotel folio in the local currency	lodging	file	complex	foreign	manual, looked up	B E G H O P
LOD-NOSHOW	No-show charge for a reservation not cancelled	lodging	not filed	simple	domestic	manual	C
LOD-CANCEL-FEE-AIRLINE	Hotel cancellation fee caused by an airline cancellation	lodging	file + flag	simple	domestic	experience, manual	K
LOD-AIRBNB-STAFF	Airbnb stay by a staff member	lodging	file + flag	complex	domestic; staff	manual	K
LOD-AIRBNB-STUDENT	Airbnb stay by a student, no prior approval in the folder	lodging	file + flag	complex	domestic; student	manual, decision	H
LOD-SUITE	Junior suite instead of a standard room	lodging	file + flag	complex	domestic	manual	G
LOD-PERSONAL-DAYS	Two personal nights added to the stay	lodging	file part	complex	domestic	manual	I
LOD-COMPANION-SURCHARGE	Additional-guest charge for the spouse on the folio	lodging	file part	complex	domestic	manual	H
LOD-DEPOSIT-DUP	Deposit receipt plus the final folio that applies it	lodging	not filed	complex	domestic	manual	G
LOD-BOOKING-CONF	Reservation confirmation with no payment taken	lodging	not filed	simple	domestic	manual	A
LOD-RESORT-FEE	Mandatory destination / resort fee on the folio	lodging	file + flag	complex	domestic	experience	J
MEAL-TRAVEL-SMALL	Travel meal alone, under \$75	meals	file	simple	domestic	manual, receipt, decision	A B C D F I J K M N
MEAL-TRAVEL-BIG-ITEMIZED	Travel meal over \$75 with the itemised check	meals	file	simple	domestic	manual	A G
MEAL-TRAVEL-BIG-SLIP	Meal over \$75 with only the card slip	meals	file + flag	simple	domestic	manual	C
MEAL-DELIVERY	Food delivery to the hotel with fees and a tip	meals	file	simple	domestic	receipt, decision	A
MEAL-BUSINESS	Business dinner with collaborators, no alcohol	meals	file	complex	domestic	manual	B I

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (4/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
MEAL-BUSINESS-ALCOHOL	Business meal with wine on the check	meals	separate out	complex	domestic	experience, manual	A P
MEAL-BUSINESS-ALCOHOL-GRANT	Business meal with wine, on a federal grant	meals	separate out	complex	domestic; grant	manual	C
MEAL-GROUP-FUNCTION	Group dinner for twelve with the invitation list	meals	file	complex	domestic	manual	B
MEAL-TIP-EXCESSIVE	Lunch with a tip near 80%	meals	file + flag	simple	domestic	experience, manual	A
MEAL-INTL-PERDIEM	Meal receipt from an international day	meals	not filed	simple	foreign	manual	B E G H O
TA-INTL-ITINERARY	Per diem for the foreign leg, claimed through Concur's Travel Allowance itinerary	meals	file	complex	foreign	experience, manual	B E G H O P
TA-INTL-PROVIDED-MEALS	Meals provided abroad (the host's lunch and dinner), deducted from the per diem	meals	file + flag	complex	foreign	experience, manual, looked up	E G
MEAL-LOCAL-PERSONAL	Lunch alone on a local (< 50 mile) business day	meals	not filed	simple	local	manual	L
MEAL-LOCAL-IN-MEETING	Lunch during an all-day local meeting	meals	file + flag	simple	local	manual	L
MEAL-LOCAL-BUSINESS	Local lunch with a visiting collaborator	meals	file	complex	local	manual	L
MEAL-CELEBRATION	Cake for a lab member's birthday	meals	not filed	simple	local	manual	L
MEAL-WITH-SPOUSE	Dinner for two where the second person is the spouse	meals	file part	simple	domestic	experience, manual	D H
MEAL-AFFIDAVIT	Meal with no receipt, covered by a Missing Receipt Acknowledgement form	meals	file + flag	simple	domestic	manual	C
MEAL-DUP-SLIP-ITEMIZED	Itemised check AND card slip for the same meal	meals	not filed	simple	domestic	manual, receipt	F
MEAL-ATTENDEE-APPROVER	Business lunch where one attendee is the report's approver	meals	file + flag	complex	domestic	manual	E
MEAL-INTERNAL-STAFF	Working lunch with two GW colleagues on the trip	meals	file	complex	domestic	manual	E
MEAL-COFFEE-SMALL	Coffee and a pastry	meals	file	simple	any	manual	A C D J K
GT-UBER	Ride-share between airport, hotel and venue	ground	file	simple	any	manual, receipt	A B D E H I J K
GT-UBER-TIP-UPDATED	Original and UPDATED ride receipts (tip added afterwards)	ground	not filed	simple	domestic	receipt, looked up	A
GT-UBER-DUP-IDENTICAL	The same ride receipt saved twice under two names	ground	not filed	simple	any	receipt	C

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (5/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
GT-TAXI	Taxi with a printed meter receipt	ground	file	simple	domestic	experience, manual	A
GT-TAXI-FOREIGN	Taxi abroad, receipt in the local currency	ground	file	complex	foreign	manual	E G O
GT-TRANSIT	Public transit fare-machine receipt	ground	file	simple	any	manual	A B J O
GT-SHUTTLE	Airport shuttle van	ground	file	simple	domestic	manual	E
GT-RAIL-ACELA-BUSINESS	Acela Business Class instead of a flight	ground	file	simple	domestic	manual, looked up	B
GT-RAIL-ACELA-FIRST	Acela First Class (upgrade from Business)	ground	file part	simple	domestic	manual, looked up	G
GT-RAIL-COACH	Northeast Regional coach ticket	ground	file	simple	domestic	manual	J
GT-BUS	Intercity bus ticket home from New York / Boston	ground	file	simple	domestic	experience, manual	J
GT-RENTAL-PREFERRED	Hertz / Enterprise / National rental, no extra insurance	ground	file	complex	domestic	manual	C P
GT-RENTAL-OTHER-CDW	Avis / Budget rental WITH CDW and liability bought	ground	file	complex	domestic; staff	manual	K
GT-RENTAL-OTHER-NOCDW	Avis / Budget rental WITHOUT the required insurance	ground	file + flag	complex	domestic; staff	manual	F
GT-RENTAL-PREFERRED-EXTRA-INSURANCE	Hertz rental with an LDW bought although insurance is included	ground	file + flag	complex	domestic	experience, manual	I
GT-RENTAL-PREPAID-FUEL	Rental with the prepaid fuel option	ground	file + flag	complex	domestic	manual	I
GT-RENTAL-FUEL	Gas station receipt for the rental car	ground	file	simple	domestic	manual	C P
GT-RENTAL-FUEL-DATE-MISMATCH	Refuel receipt dated two days AFTER the rental was returned	ground	file + flag	complex	domestic	experience, manual	F
GT-RENTAL-SPECIALTY	Premium SUV / luxury rental without approval	ground	file + flag	complex	domestic	manual	K
GT-RENTAL-INTL-STAFF	Rental abroad by staff with CDW and liability	ground	file	complex	foreign; staff	manual	B P
GT-RENTAL-INTL-STUDENT	Rental abroad by a student	ground	not filed	simple	foreign; student	manual	H
GT-RENTAL-VAN15	15-passenger van rented by a student group	ground	not filed	simple	domestic; student	manual	H
GT-RENTAL-STUDENT-NONPREFERRED	Student rents from a non-preferred company	ground	file + flag	simple	domestic; student	manual	D

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (6/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
GT-RENTAL-CARWASH	Car wash receipt	ground	not filed	simple	domestic	manual	F
GT-RENTAL-REPAIR	Tire repair receipt	ground	not filed	simple	domestic	manual	F
GT-PARKING-FINE	Parking citation	ground	not filed	simple	domestic	manual	D
GT-GAS-PERSONAL	Gas for personal car	ground	not filed	simple	domestic	manual	C
GT-RENTAL-TOLL	Toll charges while driving the rental	ground	file	simple	domestic	experience, manual	C I
GT-EZPASS-MIXED	E-ZPass statement mixing trip tolls with commute tolls	ground	file part	complex	local	manual	L
GT-MILEAGE-OFFICE	Mileage from the office to a meeting and back	ground	file	complex	local	manual, looked up	L
GT-MILEAGE-COMMUTE	Commute logged as mileage	ground	not filed	simple	local	manual	L
GT-MILEAGE-FROM-HOME	Mileage logged from home when the office is closer	ground	file part	complex	local	manual	L
GT-MILEAGE-TWO-TRAVELERS	Two colleagues in one car, both logging mileage	ground	file + flag	complex	local	manual	L
GT-MILEAGE-AIRPORT-DROPOFF	Spouse drives the traveler to the airport and back	ground	file + flag	complex	domestic	experience, manual	K
GT-CAR-IN-LIEU-OF-AIR	Drove 420 miles instead of flying, with the airfare comparison	ground	file part	complex	domestic	manual	N
GT-DRIVE-200PLUS-NO-QUOTE	Drove more than 200 miles with no cost comparison	ground	file + flag	complex	domestic	manual	M
GT-PARKING	Parking garage receipt at the meeting venue	ground	file	simple	any	manual	C M N
GT-LIMO-NOAPPROVAL	Black-car service without the VP's written pre-approval in the folder	ground	file + flag	simple	domestic	manual, decision	B
GT-LIMO-MEMO	Private sedan service with the VP's written pre-approval	ground	file	complex	domestic	manual	G
GT-STALE-STAFF	Ride receipt from a different trip, more than 60 days old (staff)	ground	not filed	simple	any; staff	manual	C
GT-STALE-STUDENT	Old receipt from an earlier trip (student — 60-day rule waived)	ground	not filed	simple	any; student	experience, manual	D
GT-PERSONAL-DETOUR	Ride to a personal destination during the trip	ground	not filed	simple	domestic	manual	F
REG-PLAIN	Conference registration	registration	file	simple	any	manual, receipt	B D H J

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (7/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
REG-MEMBERSHIP-PAID	Registration invoice with a paid society membership on it	registration	separate out	simple	any	experience, receipt, looked up	A
REG-MEMBERSHIP-COMP	Registration with a complimentary (\$0) membership line	registration	file	simple	any	receipt	M
REG-WORKSHOP	Pre-conference tutorial add-on	registration	file	simple	any	manual	A K
REG-BANQUET	Optional conference banquet ticket	registration	file + flag	simple	any	experience, manual	D
REG-PCARD	Registration paid with the GW P-Card	registration	file + flag	simple	any	manual, decision	C
REG-ABSTRACT-FEE	Abstract submission / publication fee	registration	file + flag	simple	any	experience	E
REG-MEMBERSHIP-ONLY	Society membership dues on their own	registration	file + flag	simple	any	experience, manual, decision	G
REG-VIRTUAL	Online-only conference registration	registration	file + flag	simple	any	experience, manual, decision	E
OTH-HOTEL-INTERNET-SEPARATE	Hotel Wi-Fi day pass receipt	other	file	simple	any	manual	G
OTH-PHONE-ROAMING	Mobile roaming charges	other	file + flag	simple	foreign	manual	B
OTH-POSTER	Poster printing	other	file + flag	simple	any	experience, manual	A
OTH-LUGGAGE-STORAGE	Luggage storage	other	file	simple	any	manual	C
OTH-VALET	Valet parking at the meeting venue	other	file	simple	any	manual	D I
OTH-NEWSPAPER	Newsstand receipt	other	not filed	simple	any	manual	A
OTH-BOOKS-RESEARCH	Conference proceedings volume	other	file + flag	simple	any	experience, manual	E
OTH-CARD-LATE-FEE	Credit-card late fee	other	not filed	simple	any	manual	E
OTH-CLOTHING	Clothing bought on the trip	other	not filed	simple	any	manual	F
OTH-HAIRCUT	Haircut	other	not filed	simple	any	manual	F
OTH-LUGGAGE-PURCHASE	Replacement suitcase	other	not filed	simple	any	manual	F
OTH-TOILETRIES	Pharmacy receipt (toiletries)	other	not filed	simple	any	manual	D

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (8/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
OTH-MEDICINE	Over-the-counter medicine	other	not filed	simple	any	manual	H
OTH-VACCINATION-REQUIRED	Travel clinic — required vaccination	other	file + flag	simple	any	manual	E
OTH-VACCINATION-FLU	Flu shot	other	not filed	simple	any	manual	D
OTH-GIFT-NOAPPROVAL	Flowers for the session chair	other	file + flag	simple	any	manual, decision	H
OTH-EVENT-DECOR	Table decorations for the GW-hosted reception	other	file	simple	any	manual	F
OTH-CHARITY	Donation receipt	other	not filed	simple	any	manual	H
OTH-LOST-PROPERTY	Replacement of a stolen charger	other	not filed	simple	any	manual	G
OTH-DEPENDENT-CARE	Babysitter while away	other	not filed	simple	any	manual	G
OTH-PET-CARE	Dog boarding	other	not filed	simple	any	manual	G
OTH-CLUB-FEES	Golf with the collaborators	other	file + flag	simple	any	manual, decision	A
OTH-ENTERTAINMENT-LAVISH	VIP table at a club for the collaborators, no VP approval in the folder	other	file + flag	simple	any	experience, manual, decision	G
OTH-SUPPLIES	Office supplies bought on the trip	other	file + flag	simple	any	experience	C
OTH-SPORTS-EQUIPMENT	Running shoes	other	not filed	simple	any	manual	G
OTH-ATM-FEE	ATM withdrawal abroad — the operator and foreign-transaction fees are conversion fees	other	file part	simple	foreign	manual, decision	B P
OTH-FX-FEE-STATEMENT	Card statement excerpt with foreign-transaction fees	other	file	simple	foreign	manual, decision	B
OTH-PAYMENT-INDIVIDUAL	Cash paid to a student helper, handwritten receipt	other	not filed	simple	any	manual	G
OTH-VENDOR-INVOICE	An unpaid vendor invoice (Net 30) in the folder	other	not filed	simple	any	manual	C
OTH-GIFT-MEMO	Flowers for a retiring collaborator, with the dean's written approval	other	file	complex	any	manual	J
DOC-PROGRAM	Conference program (agenda)	documents	not filed	simple	any	receipt	A B J
DOC-APPROVAL-ORPHAN	An approval memo for an expense that is not in the folder	documents	not filed	simple	any	experience	B

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E2 · The 267 situations (9/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
DOC-CARD-DETAIL-ITEMIZED	Card transaction detail WITH items (valid as a receipt for goods)	documents	file + flag	simple	any	manual	E
DOC-CARD-DETAIL-BARE	Card transaction detail WITHOUT items (not a receipt)	documents	file + flag	simple	any	manual	C
DOC-NOTES-PERDIEM-DOMESTIC	Traveler asks for per diem on a domestic trip	documents	file + flag	simple	domestic	manual	D
DOC-NOTES-PURPOSE-INADEQUATE	Notes give only 'Attend conference' as the purpose	documents	file + flag	simple	any	manual	E
DOC-DATE-VS-NOTES	Taxi receipt printed after the time the notes say the traveler left the city	documents	file + flag	simple	domestic	manual, decision	M
FUND-GRANT-ENTERTAINMENT	Entertainment receipt on a federal grant	funding	not filed	simple	domestic; grant	manual	C
FX-BOTH-CURRENCIES	Taxi card slip showing the local amount AND a dynamic-currency-conversion USD amount	foreign	file + flag	complex	foreign	experience, manual	B P
FX-HANDWRITTEN	Foreign receipt with the traveler's handwritten USD conversion	foreign	file	complex	foreign	manual	E
TIME-REG-OLD-BUT-FINE	Registration paid 100 days before the trip	timing	file	simple	any	manual	F
SUBS-JOURNAL-ANNUAL	Annual journal subscription paid on the personal card	subscriptions	file + flag	simple	any	manual, looked up, decision	T
SUBS-JOURNAL-GRANT	Same journal subscription, report charged to a federal grant	subscriptions	file + flag	simple	any; grant	looked up, decision	P
SUBS-NEWS-PERSONAL	Personal digital newspaper subscription	subscriptions	not filed	simple	any	manual, looked up	R
SUBS-MONTHLY-STALE	Three monthly subscription charges, the oldest over 60 days (staff)	subscriptions	file + flag	simple	any; staff	manual, decision	T
SUBS-MONTHLY-STALE-STUDENT	Same three monthly charges, student (60-day limit waived)	subscriptions	file + flag	simple	any; student	manual, decision	U
SUBS-INVOICE-UNPAID	Unpaid institutional subscription renewal invoice billed to GW	subscriptions	not filed	simple	any	manual, looked up, decision	T
SOFT-SAAS-ANNUAL	Annual software subscription bought with a click-through licence	software	file + flag	simple	any	looked up, decision	Q
SOFT-SITE-LICENSE-DUP	Individual MATLAB licence although GW has a site licence	software	file + flag	simple	any; faculty fulltime	experience, looked up, decision	V
SOFT-AI-TOOL	ChatGPT Plus monthly charge	software	file + flag	simple	any	looked up, decision	Q
CLOUD-AWS-MONTHLY	Paid AWS monthly invoice on the personal card	software	file + flag	complex	any	experience, manual, looked up, decision	T
CLOUD-AWS-GRANT	Same AWS invoice, federal grant	software	file + flag	complex	any; grant	manual, looked up, decision	P

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E2 · The 267 situations (10/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
CLOUD-INVOICE-UNPAID	Unpaid cloud invoice billed to the department	software	not filed	simple	any	manual, decision	U
CLOUD-PERSONAL-STORAGE	Personal iCloud+ family storage plan	software	not filed	simple	any	experience, manual	T
BOOK-RESEARCH-TEXT	Reference book bought for research or teaching	extended	file	simple	any	manual, decision	R
SUPP-STAPLES-LOCAL	Office supplies bought locally on the personal card	supplies	file + flag	simple	any	manual, looked up, decision	T
SUPP-GRANT-FEDERAL	Same Staples receipt, federal grant	supplies	file + flag	simple	any; grant	looked up, decision	U
SUPP-OVER75-SLIP	Supplies over \$75 with only the card slip	supplies	file + flag	simple	any	manual, decision	R
SUPP-MIXED-PERSONAL	One Target receipt with office supplies and toiletries	supplies	file part	simple	any	experience, manual, decision	R
SUPP-PERSONAL-SCHOOL	Staples receipt for a child's school supplies	supplies	not filed	simple	any	manual	U
SUPP-AMAZON-PERSONALCARD	Amazon.com order (personal account) shipped to the office	supplies	file + flag	simple	any; student	looked up, decision	U
SUPP-LAPTOP-SELFBUY	Laptop bought at Best Buy on the personal card	supplies	file + flag	simple	any	looked up, decision	Q
SUPP-WORKSTATION-CAPITAL	GPU workstation of \$6,450 bought personally	supplies	not filed	simple	any; faculty fulltime	experience, looked up, decision	Q
SUPP-ON-TRIP	Office supplies receipt in a trip folder (mixed report)	mixed	file + flag	simple	domestic	manual, looked up, decision	W
DUES-SOCIETY-ANNUAL	Professional society dues on their own	memberships	file + flag	simple	any	manual, looked up, decision	Q
DUES-IN-TRIP-FOLDER	Society dues receipt found in a trip folder	mixed	file + flag	simple	any	manual, looked up, decision	W
DUES-BUNDLED-SPLIT	Registration receipt with a paid membership line — true split	mixed	separate out	complex	any	manual, receipt, decision	W
DUES-PTFPDF-OVERCAP	Part-time faculty dues of \$150 under the professional-development fund	memberships	file part	complex	any; faculty parttime	experience, looked up, decision	R
DUES-FULLTIME-SAME-DUES	Same \$150 dues receipt, full-time faculty	memberships	file + flag	simple	any; faculty fulltime	looked up, decision	V
DUES-HEALTH-CLUB	Annual gym membership	memberships	not filed	simple	any	manual	T
DUES-AIRLINE-LOUNGE-ANNUAL	Annual United Club membership	memberships	not filed	simple	any; faculty fulltime	manual	V
DUES-DEPT-INVOICE-UNPAID	Unpaid society invoice for the department's institutional membership	memberships	not filed	simple	any	manual, looked up, decision	V

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E2 · The 267 situations (11/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
EVENT-FLOWERS	Flowers for a university event	events	file	simple	any	manual, decision	R
EVENT-RECEPTION-ALCOHOL	Catered university reception with wine and beer	events	separate out	complex	any	manual, decision	W
EVENT-EMPLOYEE-BIRTHDAY	Balloons and a banner for a colleague's birthday	events	not filed	simple	any	manual, decision	U
GIFT-APPROVED-MEMO	Retirement gift with the dean's written pre-approval	events	file	complex	any	manual, decision	U
GIFT-NOAPPROVAL	Same gift, no approval in the folder	events	file + flag	simple	any	manual, decision	T
EVENT-SPEAKER-HONORARIUM	Seminar speaker's honorarium paid personally by Zelle	events	not filed	simple	any; faculty fulltime	manual, decision	V
PRINT-POSTER-LOCAL	Poster and flyer printing for an on-campus event	extended	file	simple	any	experience, manual, decision	R
POST-FEDEX-SHIPMENT	Overnight shipment of samples to a collaborator	extended	file	simple	any	experience, manual, decision	Q
RECRUIT-CANDIDATE-DINNER	Dinner with a faculty candidate, no alcohol	recruiting	file	complex	any; faculty fulltime	manual, decision	V
RECRUIT-DINNER-ALCOHOL	Candidate dinner with two glasses of wine	recruiting	separate out	complex	any; faculty fulltime	experience, manual, decision	Q
RECRUIT-EMPLOYEE-PAID-HOTEL	Employee paid the candidate's hotel night on a personal card	recruiting	file + flag	simple	any	manual, decision	Q
RECRUIT-TRIP-RECRUIT-STUDENTS	Employee's own trip to recruit students — ordinary travel	recruiting	file	simple	domestic; staff	manual, decision	W
RECRUIT-JOB-AD-PERSONALCARD	Job advertisement paid personally	recruiting	file + flag	simple	any; faculty fulltime	manual, looked up, decision	Q
RECRUIT-JOB-AD-GRANT	Same job advertisement, federal grant	recruiting	file + flag	simple	any; grant; faculty fulltime	manual, looked up, decision	V
RECRUIT-CANDIDATE-SELFPAID	The candidate's own receipts and reimbursement request	recruiting	not filed	simple	any; staff	manual, looked up, decision	W
RECRUIT-HEADHUNTER-INVOICE	Search firm's fee invoice	recruiting	not filed	simple	any; staff	manual, looked up, decision	W
RECRUIT-SIGNING-BONUS	Offer letter with a signing bonus filed as 'to be reimbursed'	recruiting	not filed	simple	any; staff	looked up, decision	T
EVENT-FOREIGN-SPEAKER-AIRFARE	Employee paid a foreign guest lecturer's airfare	events	file + flag	simple	any	manual, decision	V
RELOC-FACULTY-MOVERS	New assistant professor's moving-company invoice	relocation	not filed	simple	any; faculty fulltime	experience, looked up, decision	Q
RELOC-FACULTY-HOUSEHUNT	New faculty house-hunting trip before the start date	relocation	not filed	simple	any; faculty fulltime	experience, looked up, decision	V

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (12/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
RELOC-PARTTIME-FACULTY	Part-time faculty member's U-Haul receipt	relocation	not filed	simple	any; faculty parttime	experience, looked up, decision	R
PCARD-SUBSCRIPTION	Journal subscription receipt showing the GW P-Card	pcard	file + flag	simple	any; staff	manual, decision	—
PCARD-GUEST-AIRFARE	Candidate's ticket booked as Guest Traveler on the department P-Card	pcard	file + flag	simple	any; staff	manual, looked up, decision	—
PCARD-SUPPLIES-AMAZON	Amazon Business order paid with the P-Card, tax exempt	pcard	file + flag	simple	any; staff	manual, looked up, decision	—
PCARD-PERSONAL-CHARGE	Personal purchase on the GW P-Card	pcard	not filed	simple	any; staff	experience, manual	—
SUBJ-GIFTCARDS-SELFBUGHT	Gift cards bought personally for study participants	extended	not filed	simple	any; student	experience, manual, looked up, decision	U
TEL-HOTSPOT-GRANT	Mobile hotspot data plan charged to a federal grant	extended	file + flag	simple	any; grant; staff	experience, manual, decision	P
CONF-LOCAL-REGISTRATION	Registration for a conference in Washington (no travel)	extended	file	simple	any	manual, decision	Q
SEMINAR-LOCAL-FEE	Attendance fee for a one-day seminar in Washington	extended	file	simple	any	manual, decision	Q
TRAINING-ONLINE-COURSE	Online training course the project's PI asked for	extended	file + flag	simple	any	experience, manual, decision	Q
PERMIT-DRONE-FAA-REG	FAA registration fee for three research drones	extended	file	simple	any	manual, looked up, decision	Q
EVENT-INTL-FLOWERS	Table flowers for a GW reception abroad	events	file	simple	foreign; non grant	experience, manual, decision	P
RECRUIT-CAREER-FAIR-BOOTH	Employer booth at a regional STEM career fair	recruiting	file + flag	simple	any; non grant	experience, manual, looked up, decision	P
OFFICE-NOTARY-FEE	Notary fee for a sponsor certification	supplies	file + flag	simple	any; non grant	experience, manual, looked up, decision	P
PROGRAM-DESIGN-WORKSHOP	Workshop fee to develop a new master's program	events	file + flag	simple	any; non grant	manual, looked up, decision	R
SUPP-MISC-MOVING-BOXES	Moving boxes and packing material for a lab move	supplies	file	simple	any; non grant	experience, manual, looked up, decision	Q
CONSULT-AIRFARE-EMPLOYEE-PAID	Employee bought a contracted consultant's airfare	extended	file + flag	simple	any	experience, manual, looked up, decision	Q
PROMO-ITEMS-LICENSED	Logo tote bags and stickers for a public science festival	events	file	complex	any; non grant	experience, looked up	W
PROMO-ITEMS-GRANT	Same promotional items, report on a federal grant	events	file + flag	complex	any; grant	experience, manual, looked up	U
PROMO-ITEMS-N0-APPROVAL	Same promotional items, no artwork approval in the folder	events	file + flag	simple	any; non grant	looked up, decision	Q

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (13/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
SUBJ-ADVANCE-RECONCILE	Reconciling a human-subjects cash advance	extended	file + flag	complex	any; staff	manual, looked up, decision	P
SUBJ-CASH-OWN-POCKET	Same payment log, paid from the researcher's own cash	extended	not filed	simple	any; staff	experience, manual, looked up, decision	T
ROOM-HOTEL-MEETING-ROOM	Half-day hotel meeting room for a partner workshop	extended	file + flag	simple	any	looked up, decision	P
ROOM-COWORK-MONTHLY-DESK	Monthly coworking desk for a field study	extended	file + flag	simple	any	looked up, decision	P
LEASE-COPIER-FIRST-PAYMENT	First payment of a copier lease the person signed and paid	ovpr unselectable	not filed	simple	any	manual, looked up, decision	Q
EMPLOYEE-STAFF-OUTING	Chair-approved staff recognition outing paid personally	ovpr unselectable	not filed	simple	any	manual, looked up	Q
TRAINEE-T32-SYMPOSIUM	T32 trainee's train ticket to a one-day symposium	ovpr unselectable	not filed	simple	any; staff	experience, looked up	P
SERVICE-FOREIGN-TRANSLATOR	Unpaid EUR invoice of a translator in Portugal	ovpr unselectable	not filed	simple	any	manual, looked up	Q
SERVICE-TRANSCRIPTION-SOW	Unsigned statement of work of a transcription firm	ovpr unselectable	not filed	simple	any	experience, manual, looked up	Q
CONSTRUCTION-LAB-RENOVATION	Paid deposit for renovation work in a GW lab	ovpr unselectable	not filed	simple	any	manual, looked up, decision	Q
CAPITAL-CONFERENCE-TABLE	Conference table of \$6,380 bought personally	ovpr unselectable	not filed	simple	any	looked up, decision	Q
CAPITAL-VEHICLE-DEPOSIT	Deposit on a used pickup truck for field work	ovpr unselectable	not filed	simple	any	looked up, decision	Q
CAPITAL-FABRICATION-GW-OWNED	Parts under \$5,000 each for a GW-owned instrument build	ovpr unselectable	not filed	simple	any	looked up, decision	Q
CAPITAL-FABRICATION-GOV-OWNED	Parts under \$5,000 each for a cryostat test rig on a federal contract that gives title to the Government	ovpr unselectable	not filed	simple	any	looked up, decision	P
CAPITAL-COPIER-PURCHASE	A 44-inch large-format printer of \$6,400 bought outright on a personal card	ovpr unselectable	not filed	simple	any	looked up, decision	P
CAPITAL-LAB-INSTRUMENT	Refrigerated centrifuge of \$8,450 with installation, bought personally	ovpr unselectable	not filed	simple	any	looked up, decision	Q
EQUIPMENT-GOV-OWNED	A UV-Vis spectrophotometer bought for a federal contract: government property	ovpr unselectable	not filed	simple	any	experience, looked up	P
LAB-SUPPLIES-GRANT	Laboratory consumables charged to the federal award	supplies	file	simple	any; grant	manual, looked up, decision	S
LAB-SUPPLIES-DRYICE	Dry ice and a shipper for a sample shipment	supplies	file	simple	any	manual, looked up, decision	S
LAB-BENCH-STOOLS	Four lab bench stools bought out of pocket	supplies	file + flag	simple	any; non grant	looked up, decision	S

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E2 · The 267 situations (14/14)

SITUATION	THE DOCUMENT AND ITS CIRCUMSTANCE	FAMILY	HANDLING	FORM WORK	CONDITIONS	BASIS	TRIPS
SUPP-SHREDDER-OFFICE	A cross-cut shredder for the lab's office	supplies	file	simple	any	experience, manual, looked up, decision	S
PRINT-SUPPLIES-TONER	Toner and plotter paper for the lab's printer	supplies	file + flag	simple	any	experience, looked up, decision	S
PHOTO-SUPPLIES-FIELD	Memory cards and batteries for photographing field samples	supplies	file	simple	any	experience, manual, decision	S
LIB-SUPPLIES-ARCHIVE	Archival boxes and sleeves for the project's field notebooks	supplies	file + flag	simple	any	experience, manual, decision	S
BIND-REPORT-COPIES	Three copies of the final report bound at a shop	extended	file	simple	any	experience, decision	S
LIB-DOC-DELIVERY	Document-delivery fees for three articles	extended	file	simple	any	experience, manual, decision	S
FUEL-FLEET-VAN	Fuel an employee paid for a GW fleet van	extended	file + flag	complex	any	experience, manual, looked up, decision	S

Travel first (169), then the others (98), by family. Trips: the trips that use the situation. Source: ExpenseAI facts of 2026-09-22.

E3 · The 23 trips

TRIP	REPORT NAME	TRAVELER	FUNDING	CITIES	DAYS	DOCUMENTS	LINES	NOT FILED	FLAGS	LATEST SCORE
A	MOSS 2026 Chicago	staff	Non-Grant	Chicago, Illinois	4	27	23	5	6	96.6%
B	NEDS 2026 Boston + Montreal	staff	Non-Grant	Boston, Massachusetts + Montreal, Quebec	4	26	18 + 4 per-diem days	8	7	96.6%
C	SEAM 2026 Atlanta (NSF award)	staff	Grant (federal)	Atlanta, Georgia	3	25	18	8	10	93.4%
D	WCAI 2026 San Francisco	student	Non-Grant	San Francisco, California	7	19	12	7	5	100.0%
E	ICML-East 2026 NYC + London	staff	Grant (federal)	New York, New York + London, United Kingdom	3	21	16 + 6 per-diem days	5	11	90.7%
F	PNW Stats 2026 Seattle	staff	Non-Grant	Seattle, Washington	6	21	13	8	7	93.7%
G	NEMS 2026 Boston + Paris	staff	Non-Grant	Boston, Massachusetts + Paris, France	6	21	11 + 5 per-diem days	10	7	97.0%
H	MWAI 2026 Chicago + Toronto	student	Non-Grant	Chicago, Illinois + Toronto, Ontario	6	18	10 + 4 per-diem days	8	4	92.9%
I	SEAM Follow-up 2026 Atlanta	staff	Non-Grant	Atlanta, Georgia	8	14	12	2	5	100.0%
J	NYC Data Week 2026	staff	Non-Grant	New York, New York	10	13	10	3	1	100.0%
K	BayML 2026 San Francisco	staff	Non-Grant	San Francisco, California	9	15	13	2	7	94.5%
L	Local travel Aug-Sep 2026	staff	Non-Grant	Washington, District of Columbia	3	7	6	4	3	86.2%
M	Boston site visit Sep 2026	staff	Non-Grant	Boston, Massachusetts	4	7	6	1	2	87.0%
N	NYU seminar Sep 2026	staff	Non-Grant	New York, New York	2	6	4	2	1	100.0%
O	Alan Turing Inst. visit 2026	staff	Non-Grant	New York, New York + London, United Kingdom	1	7	4 + 5 per-diem days	3	2	100.0%
P	BCSS 2026 Boston + London	staff	Non-Grant	Boston, Massachusetts + London, United Kingdom	3	29	21 + 5 per-diem days	9	24	78.5%
Q	New lab start-up Aug–Sep 2026	faculty fulltime	Non-Grant	no travel	—	30	16	15	38	89.4%
R	Fall 2026 teaching expenses	faculty parttime	Non-Grant	no travel	—	12	7	5	8	72.5%
S	Lab supplies and equipment Sep 2026	staff	Non-Grant	no travel	—	11	10	1	5	87.8%
T	Lab subscriptions and services Sep 2026	staff	Non-Grant	no travel	—	13	6	7	10	93.3%
U	PhD student purchases Sep 2026	student	Non-Grant	no travel	—	14	7	7	9	87.0%
V	Assistant professor expenses Sep 2026	faculty fulltime	Non-Grant	no travel	—	13	5	8	11	78.3%
W	Seattle student recruiting Sep 2026	staff	Non-Grant	Seattle, Washington	3	13	9	6	9	100.0%

Documents: the files the agent sees, the trip notes included. Lines: the expense lines the answer key expects. Latest score: the last valid batch of each trip, checks passed ÷ checks.
Source: ExpenseAI facts of 2026–09–22.

E4 · Terms

TERM	MEANING	EXAMPLE
policy	the rules the reports must follow: GW's travel and business expense manual, GW web pages and federal rules	manual App. B p.30–31
provision	one rule of the policy, recorded with its page or link; actionable if it decides how a document is filed	APPB-PERSONAL
situation	one case the policy decides, apart from any trip: a kind of document, its conditions, the correct handling and its basis	LOD-FOLIO-PERSONAL
conditions (scope)	where a situation applies: the kind of city, the traveler, the funding	LOD-FOLIO-INTERNET-GRANT: on a federal grant
family	the kind of expense a situation is about; 10 in travel, 11 outside it	lodging
handling	what must become of a document: file · file + flag · separate out · file part · not filed	file part
judgment call	a situation whose answer is more than copying the receipt	MEAL-BUSINESS-ALCOHOL
form work (difficulty)	how much filling the correct entry takes: simple or complex	complex: the nightly itemization
trap	a deliberate contradiction or duplicate among the documents	AIR-WIFI-OTHER-CARRIER
basis	what an answer rests on: the manual or a GW rule, our decision, a source we looked up, experience, one of our receipts	AIR-COMPANION: manual p.25
instance	one generated copy of a situation: one to three documents and their answer key	Hyatt Regency Chicago folio
answer key	what a correct report contains: lines, flags, documents left out, totals that must not appear, summary mentions	amount 1120.60; unreduced total 1182.59
expense line	one entry of a Concur report: account, amount, date, vendor, currency and the fields its account requires	53104 US lodging, 1,120.60
flag	a note the agent leaves for a person to decide before submitting; in non-travel data either a notice or needs review	the alcohol split of a business lunch
itemization	a lodging line broken down night by night, room rate and tax	four nights, 08-10 to 08-13
not in Concur	a real expense that goes through another channel, so it is not filed	a \$6,450 workstation: fixed assets
per diem	on a foreign leg, meals paid as a daily allowance instead of from receipts	trip E, London
trip	a traveler's story and its situations in one folder: one Concur report	trip A, MOSS 2026 Chicago
trip notes	the traveler's own notes in the folder: purpose, dates, funding, remarks and attendees	trip_notes.txt
turn	one instruction to the agent with a few of the trip's documents; a trip takes two to six	turn 2: the folio and the confirmation
run	one turn's execution, recorded step by step: page seen, decision, actions, screenshot	trip A: 4 runs, 212 steps
export	the report read back from Concur after a run	actual.json
score	checks passed ÷ checks in the answer key, reported by dimension	trip A: 84 of 87, 96.6%
ledger	each situation's record across batches: tested how often, passed how often	LOD-FOLIO-PERSONAL: 3/3

E5 · The policy’s sources (1/4)

SOURCE	WHAT IT IS	LINK
manual	GW Travel, Entertainment and Business Expense Reimbursement Manual, rev. 2024-02-21: the page numbers on the data pages	ibuy.gwu.edu
gw-new-manual	GW Travel & Business Expense Reimbursement Manual, Rev. 9-09-26 (the pages quoted as 'new p.N')	ibuy.gwu.edu/sites/g/files/zaxdzs4686/files/2026-09/Travel_B
gw-expense-guidelines	iBuy+ Expense Reporting Helpful Guidelines: travel type Domestic / International / No Travel ('a combination of Domestic and No Travel? Either selection is great'); the expense types that route to OVPR on sponsored funds	ibuy.gwu.edu/expense-reporting-helpful-guidelines
gw-nonemployee-reimbursement	Reimbursements to a person who is not an employee or student: Concur Invoice, invoice type 'Non-employee reimbursement'	ibuy.gwu.edu/reimbursements-person-not-employee-or-student
gw-travel-guidelines	iBuy+ Travel Helpful Guidelines: a visitor is booked as 'Guest Traveler'	ibuy.gwu.edu/travel-helpful-guidelines
gw-office-supplies	iBuy+ office supplies and furniture: the contract suppliers	ibuy.gwu.edu/office-supplies-and-furniture
gw-amazon-business	GW's Amazon Business account: P-Card only	ibuy.gwu.edu/amazon-business-account
gw-contract-guide	Contract Process Guide (Procurement, 2025-04-30): iBuy+ is the preferred purchasing method (p.8); the order of purchasing methods (p.7); security assessment (p.6–7); click-through terms are contracts (p.33–35); the competitive-exemptions list (App. A p.48–50)	procurement.gwu.edu/sites/g/files/zaxdzs4691/files/2025-05/C
gw-exemptions	Competitive exemptions list: dues (12), subscriptions (13), software subscription renewals (15), advertising (30/32), search firms (42) — payable by P-Card or payment request, no purchase order needed	procurement.gwu.edu/appendix-procurement-competitive-exempti
gw-procurement-methods	Procurement methods: 'All software and SaaS purchases, regardless of dollar amount, must have a security assessment'	procurement.gwu.edu/procurement-methods
gw-pcard-overview	P-Card overview and benefits	procurement.gwu.edu/overview-benefits
gw-cardholder	P-Card cardholder responsibilities	procurement.gwu.edu/cardholder-responsibilities
gw-direct-charging	Direct charging of administrative costs to sponsored projects: subscriptions, computers, general office supplies, memberships, postage	compliance.gwu.edu/direct-charging-administrative-costs-asso
gw-standardization	Standardization of goods, equipment and services	compliance.gwu.edu/standardization-goods-equipment-and-servi
gw-it-risk	GW IT cybersecurity risk assessments: when one is required	it.gwu.edu/cybersecurity-risk-assessments
gw-ai-tools	GW IT: status of AI tools — the approved list	it.gwu.edu/ai-tools-status
gw-ai-guidance	GW IT: AI guidance and best practices — use only tools approved through procurement and risk review	it.gwu.edu/ai-guidance-and-best-practices
gw-workstation	Faculty/staff workstation guidelines: 'All computer purchases must be ordered through GW IT'	it.gwu.edu/faculty-staff-workstation-initiative-guidelines
gw-research-cloud	GW IT Research Cloud: self-service AWS resources for researchers	it.gwu.edu/research-cloud

The manual, and the 60 web sources registered for the expenses outside travel. Source: ExpenseAI benchmark/pool/coverage_newcats.py.

E5 · The policy’s sources (2/4)

SOURCE	WHAT IT IS	LINK
gw-capitalization	Controller: capitalized asset management — \$5,000 unit price; 58524 capitalized, 52193 / 52199 not	controller.gwu.edu/capitalized-asset-management
gw-osp-budget	Sponsored projects: budget cost categories	sponsoredprojects.gwu.edu/budget-cost-categories
gw-osp-expenditures	Sponsored projects: 'alcohol, entertainment, advertising, and proposal costs are generally not allowable'	sponsoredprojects.gwu.edu/award-expenditures
gw-site-licenses	SEAS software licenses: MATLAB, Microsoft Office and other site licences	engineering.gwu.edu/software-licenses
gw-aaas	GW is an AAAS institutional member: the Elemental membership is free to faculty, postdocs and students	research.gwu.edu/aaas-elemental-memberships
gw-free-news	GW Today 2020-11-09: free digital New York Times, Washington Post and Wall Street Journal for students, faculty and staff	gwtoday.gwu.edu/students-leadership-collaborate-expand-free-
gw-ptfpdf	Part-time Faculty Professional Development Fund: 'Up to one hundred dollars of membership for a professional organization may be reimbursed'	facultyaffairs.gwu.edu/professional-development-fund
gw-relocation	Relocation and moving allowance for new faculty: a lump sum in the normal paycheck; none for part-time or one-semester faculty	facultyaffairs.gwu.edu/relocation-and-moving-allowance-new-f
gw-faculty-handbook	GW Faculty Handbook (2025) §2.3 Transition Assistance for New Full-Time Faculty: \$3,000 / \$4,500, taxable	provost.gwu.edu/sites/g/files/zaxdzs5926/files/2025-07/2025%
gw-hr-why-gw	HR 'Why GW?': third-party moving discounts only — no staff relocation policy	hr.gwu.edu/why-gw
gw-hr-compensation	HR Managing Compensation: signing bonuses	hr.gwu.edu/managing-compensation
gw-hr-special	HR special circumstances: external search firms	hr.gwu.edu/special-circumstances
gw-hr-advertising	HR advertising chart (2024-06): fee-based options are department / school / division funded	hr.gwu.edu/sites/g/files/zaxdzs5691/files/2024-06/gw-adverti
gw-ad-placement	Advertising placement policy: who coordinates which advertisement	compliance.gwu.edu/advertising-placement
cfr-200-454	2 CFR 200.454 Memberships, subscriptions, and professional activity costs	www.law.cornell.edu/cfr/text/2/200.454
cfr-200-463	2 CFR 200.463 Recruiting costs	www.law.cornell.edu/cfr/text/2/200.463
cfr-200-464	2 CFR 200.464 Relocation costs of employees	www.law.cornell.edu/cfr/text/2/200.464
gw-unallowable-costs	Identification and Treatment of Unallowable Costs on Sponsored Projects Policy: the accounts excluded from federal claims — 52612 and 55152 among them, 52722 but not 52721 (the '-policy' URL of the brief returns 404)	compliance.gwu.edu/identification-and-treatment-unallowable-
gw-mro-supplies	iBuy+ Facilities and MRO supplies: Home Depot is one of GW's contracted suppliers	ibuy.gwu.edu/facilities-and-mro-supplies
gw-hotel-event-requests	Procurement: Hotel & Event Requests — hotel meeting and event space sourced through a CVENT request; the assigned buyer executes and releases the PO; invoices go to University Payables (no dollar threshold stated)	procurement.gwu.edu/gw-hotel-event-requests

The manual, and the 60 web sources registered for the expenses outside travel. Source: ExpenseAI benchmark/pool/coverage_newcats.py.

E5 · The policy’s sources (3/4)

SOURCE	WHAT IT IS	LINK
gw-finance-departments	Finance Division departments: the Office of Real Estate sources, negotiates and administers off-campus space leases	finance.gwu.edu/departments
gw-merchandise-guidelines	Licensing & Trademark Program, departmental merchandise guidelines (undated, 2019–20): items with GW's marks come from a GW-licensed vendor; the artwork is approved first (go.gwu.edu/merchapproval)	business-services.gwu.edu/sites/g/files/zaxdzs6001/files/dow
gw-cash-advance-request	iBuy+ Expense: Requesting a Cash Advance (2019) — requested by the employee, approved by the department and Procurement, deposited to the employee's payroll account	ibuy.gwu.edu/sites/g/files/zaxdzs4686/files/downloads/EXP%20
gw-cash-advance-apply	iBuy+ Expense: Cash Advances — Adding to a Report and Returning a Balance (Aug 2024): 'ALL CASH ADVANCES MUST BE APPLIED TO AN EXPENSE REPORT AND SUBMITTED'	ibuy.gwu.edu/sites/g/files/zaxdzs4686/files/2024-08/EXP%20Ca
gw-forte-sop	Standard Operating Procedures for the Forte (Advarra) Participant Payments System (May 2024): Advarra replaces gift cards and their expense reports; another method is set with the Finance Division and noted in the IRB application	procurement.gwu.edu/sites/g/files/zaxdzs4691/files/2024-05/f
gw-procure-pay-resources	Procure-to-Pay resources: 'The Advarra Participant Payments solution is the university's preferred payment option for research study participants'	procurement.gwu.edu/procure-pay-resources
gw-procurement-manual	GW Procurement Manual, Version 6.1 (June 13, 2024), the PDF the Procurement Manual page links: p.8 new suppliers register in iSupply, invoices go to University Payables, 'Do not pay for services from international individuals on a p-card'; p.34 §6.1.2 only delegated individuals sign contracts, §6.2 capital and operational leases (the specification read v6.0 of 2024-04-16, where §6.1.2 and §6.2 were on p.33)	procurement.gwu.edu/media/2141
gw-mfd-suppliers	P2P multi-function device suppliers: every new printer or copier comes from one of two contracted vendors, requested on the order placement form through Gatekeeper; a purchase option exists	procurement.gwu.edu/multi-function-device-suppliers
gw-international-transactions	Procurement: international transactions — the payee registers with a W-8BEN / W-8BEN-E; an approved invoice, no PO needed, EFT recommended	procurement.gwu.edu/international-transactions
gw-worker-classification	Tax Department: worker classification — an independent contractor is classified first and paid through University Payables	taxdepartment.gwu.edu/worker-classification
gw-worker-classification-exemptions	Tax Department: worker classification and international engagement request form exemptions — language services, translation included	taxdepartment.gwu.edu/worker-classification-and-internationa
gw-tax-employee-awards	Tax Department: processing gifts, prizes and awards to employees — charged to account 52615; non-taxable ones paid or reimbursed by University Payables through the Payment Request Form; 57121 is not for employees	taxdepartment.gwu.edu/processing-gifts-prizes-and-awards-emp
gw-hr-recognition	HR recognition: On-the-Spot Awards may be group events or outings, up to \$100 per individual per year	hr.gwu.edu/recognition-and-positive-vibes

The manual, and the 60 web sources registered for the expenses outside travel. Source: ExpenseAI benchmark/pool/coverage_newcats.py.

E5 · The policy’s sources (4/4)

SOURCE	WHAT IT IS	LINK
gw-taxation-employee-gifts	Taxation of gifts, prizes and awards to employees: de minimis non-cash gifts up to \$100; departments generally may not use the P-Card for awards	compliance.gwu.edu/taxation-gifts-prizes-and-awards-employee
gw-fixed-asset-policy	Fixed Asset Management Policy: fixed assets are bought on a PO in EAS or through iBuy+; a fabrication (a unique asset over \$5,000 built from parts, some or all under \$5,000) is reported to the Fixed Assets Department; government property defined	compliance.gwu.edu/fixed-asset-management
gw-fleet	Transportation and Logistics: fleet — the assignment, use and replacement of university vehicles; a shared motor pool of vans and trucks	transportation.gwu.edu/fleet
gw-fleet-rental	Transportation and Logistics: fleet rental — a unit reserves a vehicle in the Fleet Commander system; 'Payment for the final costs incurred during the rental period will be processed by an interdepartmental assessment via a journal entry'. The page says nothing about fuel or refuelling (read 2026-09-20)	transportation.gwu.edu/fleet-rental
gw-lab-supplies	iBuy+ Lab Supplies & Equipment: the category page of GW's contracted laboratory suppliers (the page carries the heading and the navigation only — no supplier list, no procedure, read 2026-09-20)	ibuy.gwu.edu/lab-supplies-equipment-0
faa-drone-registration	FAA: register a drone under Part 107 — \$5 a drone, valid three years	www.faa.gov/uas/getting_started/register_drone
nih-gps-trainee-costs	NIH Grants Policy Statement 11.3.8: allowable and unallowable costs of training grants — trainee travel (11.3.8.5), the trainee appointed at the time of travel	grants.nih.gov/grants/policy/nihgps/HTML5/section_11/11.3.8
far-52-245-1	FAR 52.245-1 Government Property: under a cost-reimbursement contract title to purchased property vests in the Government on delivery, to fabricated property on issuance, processing or reimbursement (e)(3)	www.acquisition.gov/far/52.245-1
cfr-200-421	2 CFR 200.421 Advertising and public relations: costs of promotional items and memorabilia are unallowable	www.law.cornell.edu/cfr/text/2/200.421

The manual, and the 60 web sources registered for the expenses outside travel. Source: ExpenseAI benchmark/pool/coverage_newcats.py.

E6 · Trip A's answer key: the expected lines

DOCUMENT	ACCOUNT	AMOUNT	SITUATION	HANDLING
American Airlines DCA-ORD receipt.pdf	53101-US AIRFARE EXPENSE	501.40	AIR-COACH-DIRECT	file
American Airlines checked bag receipt.pdf	53111-US OTHER TRAVEL AND RELATED EXPENSE	35.00	AIR-BAG	file
American Airlines seat assignment receipt.pdf	53111-US OTHER TRAVEL AND RELATED EXPENSE	34.00	AIR-SEAT-ECONOMY	file + flag
American Airlines ticket change receipt (later return).pdf	53101-US AIRFARE EXPENSE	108.00	AIR-CHANGE-REASON	file
Hyatt Regency Chicago folio.pdf	53104-US LODGING EXPENSE	1120.60 not 1182.59	LOD-FOLIO-PERSONAL	file part
Quartino Ristorante check.pdf	53102-US BUSINESS MEALS	39.27	MEAL-TRAVEL-SMALL	file
Quartino Ristorante check (2).pdf	53102-US BUSINESS MEALS	42.49	MEAL-TRAVEL-SMALL	file
RPM Steak dinner check.pdf	53102-US BUSINESS MEALS	91.53	MEAL-TRAVEL-BIG-ITEMIZED	file
The Purple Pig business lunch check.pdf	53102-US BUSINESS MEALS	94.07	MEAL-BUSINESS-ALCOHOL	separate out
The Purple Pig business lunch check.pdf	52611-ENTERTAINMENT	41.84	MEAL-BUSINESS-ALCOHOL	separate out
Uber Eats Lou Malnati's Pizzeria.pdf	53102-US BUSINESS MEALS	40.46	MEAL-DELIVERY	file
RPM Steak lunch check (large tip).pdf	53102-US BUSINESS MEALS	70.61	MEAL-TIP-EXCESSIVE	file + flag
Intelligentsia Coffee — Millennium Park coffee.pdf	53102-US BUSINESS MEALS	12.02	MEAL-COFFEE-SMALL	file
Uber Aug 10 920 AM.pdf	53103-US LOCAL TRANSPORTATION	59.90	GT-UBER	file
Uber Aug 11 955 PM.pdf	53103-US LOCAL TRANSPORTATION	20.10	GT-UBER	file
Uber Aug 12 348 PM.pdf	53103-US LOCAL TRANSPORTATION	20.85	GT-UBER	file
Uber Aug 11 UPDATED receipt.pdf	53103-US LOCAL TRANSPORTATION	21.35	GT-UBER-TIP-UPDATED	not filed
Flash Cab Company receipt.pdf	53103-US LOCAL TRANSPORTATION	52.80	GT-TAXI	file
CTA fare receipt.pdf	53103-US LOCAL TRANSPORTATION	15.00	GT-TRANSIT	file
MOSS Association registration receipt (with membership).pdf	53251-CONFERENCE/SEMINAR/TRAINING EXPENSE	590.00	REG-MEMBERSHIP-PAID	separate out
MOSS Association tutorial add-on receipt.pdf	53251-CONFERENCE/SEMINAR/TRAINING EXPENSE	90.00	REG-WORKSHOP	file
Poster printing.pdf	53111-US OTHER TRAVEL AND RELATED EXPENSE	125.50	OTH-POSTER	file + flag
Golf with the collaborators.pdf	52611-ENTERTAINMENT	315.00	OTH-CLUB-FEES	file + flag

23 expected lines of trip A, MOSS 2026 Chicago (testing_cases/trip_a_chicago/expected.json).

E6 · Trip A's answer key: documents left out, flags, and the score

DOCUMENT LEFT OUT	WHY	SITUATION	SCORE.TXT IN FULL · BATCH 20260914-02
Hyatt Regency Chicago reservation confirmation.pdf	a reservation confirmation is not a receipt — no payment was taken	LOD-BOOKING-CONF	— per-expense — ✓ American Airlines DCA-ORD receipt.pdf 501.40 USD ✓ American Airlines checked bag receipt.pdf 35.00 USD ✓ American Airlines seat assignment receipt.pdf 34.00 USD ✓ American Airlines ticket change receipt (lat 108.00 USD ✓ Hyatt Regency Chicago folio.pdf 1120.60 USD ✓ Quartino Ristorante check.pdf 39.27 USD ✓ Quartino Ristorante check (2).pdf 42.49 USD ✓ RPM Steak dinner check.pdf 91.53 USD ✓ The Purple Pig business lunch check.pdf 94.07 USD ✓ The Purple Pig business lunch check.pdf 41.84 USD ✓ Uber Eats Lou Malnati's Pizzeria.pdf 40.46 USD x RPM Steak lunch check (large tip).pdf 70.61 USD MISSING ✓ Intelligentsia Coffee — Millennium Park coff 12.02 USD ✓ Uber Aug 10 920 AM.pdf 59.90 USD x Uber Aug 11 955 PM.pdf 20.10 USD MISSING ✓ Uber Aug 12 348 PM.pdf 20.85 USD ✓ Uber Aug 11 UPDATED receipt.pdf 21.35 USD ✓ Flash Cab Company receipt.pdf 52.80 USD ✓ CTA fare receipt.pdf 15.00 USD ✓ MOSS Association registration receipt (with 590.00 USD ✓ MOSS Association tutorial add-on receipt.pdf 90.00 USD ✓ Poster printing.pdf 125.50 USD ✓ Golf with the collaborators.pdf 315.00 USD
Uber Aug 11 receipt.pdf	superseded by the updated receipt for the same ride	GT-UBER-TIP-UPDATED	
Newsstand receipt.pdf	newspapers and magazines are personal	OTH-NEWSPAPER	— dimensions — coverage 21/23 expense_type 21/21 basic_fields 21/21 itemization 1/1 attendees 1/1 skip_respected 5/5 exclusions 1/1 flags_required 5/6 comments_in_concur 6/6 summary_mentions 2/2 travel_allowance n/a stray_lines 1 blocking_errors 0 submittable True
Conference program.pdf	reference material, not an expense	DOC-PROGRAM	
trip_notes.txt	reference material, not an expense	notes	△ FLAG MISSING: AIR-SEAT-ECONOMY — expected a flag mentioning one of ['seat'] △ 1 line(s) in the report match no expected line: 71.77 53102-US BUSINESS MEALS
FLAG REQUIRED	MUST MENTION ONE OF	WHY	
AIR-SEAT-ECONOMY	seat	p.10 allows advance seat assignment within economy; Appendix B lists window/aisle seat fees as unallowable	OVERALL: 96.6% submittable: True
MEAL-BUSINESS-ALCOHOL	alcohol, wine, 52611, entertainment	allocation of tax and tip to the alcohol line	
MEAL-TIP-EXCESSIVE	tip, gratuity, excessive		
REG-MEMBERSHIP-PAID	membership, dues		
OTH-POSTER	poster, printing, type		
OTH-CLUB-FEES	group event, preauthoriz, pre-authoriz, golf, approval	golf is allowable only as part of a pre-authorised group event; no pre-authorisation in the folder — ask for it	

E7 · Where every number on the data pages comes from

NUMBER	FIELD OR COMPUTATION	FILE
160 provisions; 149 actionable	provisions.travel / non_travel: total, actionable	facts.json
267 situations: 169 travel, 98 non-travel	situations.totals	facts.json
handling 71 · 97 · 6 · 15 · 78; 196 judgment calls	situations.judgment.all (196 = 267 – 71, calc.)	facts.json
form work: simple 204, complex 63	situations.difficulty.all	facts.json
21 families: 10 travel, 11 non-travel	situations.families	facts.json
conditions: city, traveler, funding counts	situations.scope	facts.json
traps 19	situations.with_trap.all	facts.json
basis: 228 · 104 · 73 · 64 · 19	situations.basis_kinds.situations_with_kind.all	facts.json
801 instances, 903 files, 31 layouts	pool.travel / non_travel; pool.templates	facts.json
12 cities, 6 airlines, 22 people, 11 conferences	world	facts.json
23 trips: 382 documents, 257 lines, 134 not filed, 192 flags	trips.subtotals (all 23)	facts.json
all 169 travel situations used; 94 of 98 others	situations.trip_coverage	facts.json
trip A: 84 of 87 checks, 96.6%	the dimensions of score.txt, summed (calc.)	trip_a/score.txt
trip A: 4 turns, 212 steps, \$7.22, 24 minutes	the four “turn N → done” lines of the batch log, summed (calc.)	trip_a/log.txt
the folio’s line, nights and totals	expenses[situation = LOD-FOLIO-PERSONAL]	trip_a/expected.json
LOD-FOLIO-PERSONAL passed 3/3	its row	LEDGER.md
pass counts by handling: 243/253 · 84/134 · 7/10 · 33/39 · 115/126	passed/tested of each travel situation, summed by its handling (calc.)	LEDGER.md + facts.json

Files in slides/part3-data/expenseai/, copied by snapshot.py from the ExpenseAI repository (commit ee98f59); facts.json is ExpenseAI’s facts file of 2026-09-22, made by its scripts/deck_facts.py.